

Project Title:

K9MATCH: Ethical Canine Matching

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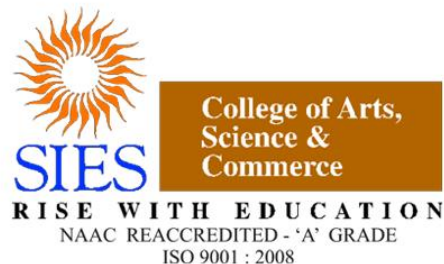
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PROJECT REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE



**S.I.E.S College of Arts, Science and Commerce,
Sion (W), Mumbai – 400 042**

K9MATCH (Ethical Canine Matching)

**For the Partial Fulfilment for The Degree of
Bachelor of Science (Computer Science) 2026-2027**

**SUBMITTED BY
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Declaration by Student

I, **Yash Prakash Patil**, hereby solemnly declare that the project report entitled “**K9MATCH: Ethical Canine Matching**”, submitted by me to the **University of Mumbai**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Science in Computer Science**, is an original work carried out by me under the guidance of **MR. RAJESH YADAV**.

I further declare that this project has been prepared solely by me. All sources of information, data, and references used in the preparation of this report have been duly acknowledged. The analysis, results, and findings presented herein are true to the best of my knowledge and belief, and no portion of this work has been copied or plagiarized from any unauthorized source.

This project represents my own intellectual contribution and independent effort towards fulfilling the academic requirements of the course. I take full responsibility for the contents of this report and confirm that it is my own genuine work, completed to fulfill the requirements of the **Bachelor of Science in Computer Science degree**.

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Abstract

The contemporary domestic pet care sector in urban India is experiencing unprecedented growth in companion animal adoption. However, canine mating and pedigree management remain dominated by informal, unverified channels like unmoderated social media groups and generic online classifieds. This informal ecosystem is plagued by severe ethical and biological hazards, including accidental inbreeding, fraudulent pedigree certificates, unregulated commercial puppy mills, and contractual disputes regarding stud fees and puppy sharing.

To address these systemic deficits, this project introduces K9MATCH: Ethical Canine Matching, an ethical, full-stack canine matchmaking, pedigree certification, and localized pet healthcare networking platform engineered using the Django (Python) framework under the Model-View-Template (MVT) architecture. K9MATCH establishes a secure, regulated digital environment where dog owners and verified breeders can discover biologically compatible mating partners, negotiate fair breeding terms, and access verified veterinary care.

The core matchmaking engine implements the spherical trigonometric Haversine Distance Algorithm mapped to an extensive database of Indian cities and OpenStreetMap Nominatim reverse-geocoded spatial coordinates. This enables precise, radius-bounded partner discovery while strictly enforcing non-negotiable biological guardrails: algorithmic prevention of self-matching, mandatory opposite-gender breeding validation, and pedigree lineage categorization. Animal welfare is assured through cryptographic two-factor authentication, administrative moderation, automated PDF generation of health passports and breeding agreements, and an integrated veterinary directory.

Keywords: Canine Matchmaking, Ethical Dog Breeding, Haversine Distance Formula, Pedigree Verification, Kennel Club of India (KCI), Django MVT, Model Validation, Veterinary Directory, Digital Pet Passport, Automated Breeding Contract.

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Chapter 1: Introduction

1.1 Background of Study:

Over the past decade, rapid urban economic expansion, dual-income households, and shifting social demographics across metropolitan India have triggered an extraordinary rise in companion animal ownership. Dogs, traditionally kept for security, have seamlessly transitioned into cherished family members ("pet parenting"). According to pet care surveys and veterinary demographic reports, India represents one of the fastest-growing pet care markets globally, registering double-digit compound annual growth rates.

Despite this affectionate cultural shift, the foundational infrastructure governing canine breeding, pedigree management, and reproductive healthcare remains deeply entrenched in an unorganized, informal, and largely unregulated black-market economy. Dog owners seeking compatible mating partners for their companion animals are confronted with immense friction. Traditionally, pet parents rely on informal word-of-mouth recommendations, unverified local pet shop brokers, or unmoderated classifieds on platforms like OLX, Quikr, Facebook Groups, and WhatsApp broadcast channels.

This informal paradigm introduces several fundamental biological, legal, and operational limitations when evaluated under animal welfare frameworks:

- **Genetic Deterioration and Inbreeding:** The absence of verified ancestral lineage documentation leads to widespread accidental inbreeding, amplifying hereditary congenital deformities such as canine hip dysplasia, brachycephalic airway obstructive syndrome, dilated cardiomyopathy, and progressive retinal atrophy.
- **Exploitative Commercial Puppy Mills:** Commercial mass breeders routinely exploit unregulated classifieds to operate illegal puppy mills, subjecting female dogs to continuous, inhumane reproductive cycles without mandatory recuperative intervals.
- **Absence of Verifiable Pedigree Grounding:** Thousands of pet parents are deceived by forged Kennel Club of India (KCI) certificates and false purebred assurances, with no centralized digital verification mechanism to validate microchips or official kennel registrations.
- **Geographic Friction and Travel Constraints:** Transporting canines over long distances for prospective mating without verified preliminary compatibility induces acute physiological distress, heatstroke, behavioral hostility, and substantial financial waste.

- **Contractual Ambiguity and Dispute Vulnerability:** Informal verbal mating arrangements routinely deteriorate into acrimonious disputes regarding stud fees, puppy-sharing rights ("pick of the litter"), veterinary liability for infectious transmission (e.g., Canine Brucellosis, Parvovirus), and post-natal care responsibility.
- **Disconnected Healthcare Ecosystems:** Owners lack integrated access to qualified nearby veterinary reproductive specialists capable of performing pre-breeding health screenings, artificial insemination, or estrus cycle ovulation tracking.

To mitigate these systemic deficits, K9MATCH introduces a structured architectural framework combining spatial algorithms, multi-tier pedigree verification, automated legal contracting, and integrated veterinary directories, thereby anchoring canine mating decisions to authenticated, welfare-compliant records.

1.2 Problem Definition:

Companion animal owners and ethical dog breeders require platforms that deliver verified, biologically compatible mating matches without exposing dogs to the risks of commercial puppy farming, genetic inbreeding, and fraudulent pedigree certificates. While commercial classifieds provide broad geographic reach, they introduce predatory commercial exploitation, provide no health or lineage vetting, and offer zero biological compatibility guardrails. Conversely, traditional kennel clubs operate via manual, paper-based studbooks that lack spatial radius discovery, instant communication, and automated contractual agreements.

Therefore, the core problem addressed by this project is: To design, implement, and deploy an ethical canine matchmaking, pedigree verification, and pet healthcare networking web platform (K9MATCH) that delivers verified, biologically sound mating matches; supports proximity-based partner discovery via spherical trigonometry (Haversine formula); enforces strict biological guardrails (opposite-gender validation and self-match elimination); ensures document authenticity through an administrative vetting pipeline; provides permission-isolated real-time chat with message management; automatically synthesizes legally binding breeding contracts and canine health passports; and incorporates localized veterinary clinic discovery with an algorithmic estrus cycle calculator.

1.3 Objectives of the Project:

To address the identified problems comprehensively, the technical objectives of K9MATCH are defined across six architectural dimensions:

- **1. Core Web Platform & Role-Based Architecture:** Design and implement a responsive, multi-tier web application using Django, PostgreSQL, and modern semantic HTML5/CSS3 supporting role-based access for dog owners, registered breeders, and platform administrators.
- **2. Geospatial Proximity Matchmaking Engine:** Implement the spherical trigonometric Haversine formula mapped to an Indian city geographic coordinate repository, enabling users to search for compatible mating partners within user-defined radial distances (e.g., 5 km to 100+ km).
- **3. Deterministic Biological Guardrails:** Enforce strict, deterministic biological validations in Django models: (a) mandatory opposite-gender biological pairing (male + female only); (b) algorithmic self-match elimination; (c) minimum breeding age enforcement (18 months); and (d) lineage classification (purebred vs. crossbreed).
- **4. Pedigree Verification & Document Moderation:** Create an administrative vetting pipeline where canine profiles requiring KCI pedigree claims remain in 'Pending Approval' until an administrator inspects uploaded certificates, microchip IDs, and vaccination records.
- **5. Negotiation Lifecycle & Direct Messaging:** Develop an atomic two-way match request negotiation pipeline (pending, accepted, rejected) that gates direct chatrooms strictly to mutually accepted pairings, featuring live message editing and retraction.
- **6. Automated Legal & Health Document Synthesis:** Integrate the ReportLab PDF Canvas graphics engine to automatically compile legal, standardized Canine Breeding Agreement Contracts and printable Canine Health Passports directly from database records.
- **7. Clinical Healthcare Networking & Estrus Tooling:** Incorporate an interactive geocoded Veterinary Directory for emergency clinic discovery and an algorithmic Canine Estrus (Heat) Cycle Calculator to assist owners in planning optimal breeding and whelping windows.

1.4 Scope of the Project:

The functional and operational boundaries of K9MATCH are defined as follows:

- **In-Scope Capabilities:** The platform encompasses the following core functional domains:
 - **User Authentication & RBAC:** Dual-role registration (Owner/Breeder) protected by Email-based OTP verification and Google OAuth 2.0.
 - **Canine Profile Management:** Rich profiles including biological attributes, multi-photo gallery uploads (up to 5MB, JPG/PNG), health badges, and behavioral ratings.
 - **Proximity Filtering:** Real-time Haversine distance computation across Indian cities with an interactive radius slider and OpenStreetMap Nominatim reverse geocoding.
 - **Two-Way Negotiation & Chat:** Dedicated, permission-isolated chatrooms with live message editing, soft deletion, and status notifications.
 - **Automated PDF Generation:** Dynamic server-side synthesis of legal Breeding Contracts and Canine Health Passports.
 - **Admin Vetting Dashboard:** Centralized dashboard for approving/rejecting listings and investigating scam reports.
 - **Healthcare Tooling:** Location-aware veterinary directory with 24/7 emergency tags and algorithmic estrus cycle estimation.
- **Out-of-Scope Capabilities (Explicit System Boundaries):** To maintain architectural focus and regulatory compliance, the platform explicitly excludes:
 - **Direct Payment Processing:** The platform facilitates stud fee negotiation but intentionally omits an integrated escrow payment gateway to avoid financial liability; transactions are completed directly between owners.
 - **Physical Transportation and Escort Logistics:** Physical vehicle dispatch, dog transit handling, and boarding logistics are outside current scope and handled privately by owners.
 - **IoT Hardware Biometric Wearables:** Real-time physiological telemetry (e.g., smart collar heart-rate or GPS tracking) is outside the web application boundary.
 - **Automated DNA Genotyping Laboratory Integration:** Direct laboratory API integration for DNA SNP marker testing is excluded; verification relies on KCI registration records and veterinary documentation.

1.5 Motivation

The introduction of modern web technologies and location-based services has revolutionized human networking and logistics. However, domestic animal welfare systems have failed to keep pace, leaving pet breeding in an opaque and ethically fraught state.

An analysis of the prevailing domestic pet breeding ecosystem in India reveals several critical concerns that motivated the architectural design of K9MATCH:

- **Combating Unregulated Puppy Mills:** Commercial puppy farming thrives on anonymity. By enforcing mandatory identity verification, vaccination tracking, and admin document vetting, K9MATCH strips unethical breeders of the cloak of anonymity, directly supporting the mandates of the Animal Welfare Board of India and the Prevention of Cruelty to Animals (Dog Breeding and Marketing) Rules.
- **Eradicating Hereditary Inbreeding Depressions:** Companion dogs deserve sound genetic health. Giving pet parents access to transparent ancestral pedigrees, breed classifications (purebred vs. crossbreed), and registered partner choices helps eradicate preventable hereditary defects that cause lifelong animal suffering and exorbitant veterinary bills.
- **Eliminating Scams and Financial Fraud:** Unregulated online classifieds are flooded with fraudulent stud listings, stolen pedigree photos, and extortionate upfront mating deposits. K9MATCH protects pet owners through verified profiles, transparent mating terms, and legally structured written contracts.
- **Promoting Responsible Pet Parenthood:** Bringing puppies into the world is a serious ethical responsibility. K9MATCH empowers dog owners with pre-breeding health checklists, heat cycle ovulation trackers, emergency veterinary directories, and digital health passports, ensuring that every breeding decision is executed with medical prudence, mutual consent, and legal clarity.

These technical and ethical imperatives motivated the development of K9MATCH as an open, secure, and compassionate canine matchmaking web platform. K9MATCH resolves these industry gaps by combining geospatial mathematical algorithms, multi-tier pedigree verification, automated contract compilation, and unified veterinary networking into a cohesive digital sanctuary.

Chapter 2: Literature Review

2.1 Existing Systems / Related Works:

The rapid growth of the companion pet industry has made partner discovery, pedigree tracking, and reproductive healthcare a critical challenge for pet parents and ethical breeders. While modern web applications offer sophisticated marketplace capabilities, utilizing generic platforms directly for canine breeding exposes companion animals to three fundamental hazards:

- **Absence of Biological Validation:** Generic classifieds platforms lack biological validation, allowing invalid mating requests (such as same-sex mating or self-matching) and failing to enforce reproductive age limits.
- **Rampant Pedigree Fraud:** Unmoderated platforms permit users to publish unverified claims regarding KCI certification and champion bloodlines, facilitating fraudulent breeding transactions.
- **Total Lack of Legal Documentation:** Over 95% of domestic dog breeding transactions occur through unrecorded verbal promises, leaving dog owners legally unprotected during disputes over stud fees, litter ownership, or medical liability.

To resolve these deficits, specialized matchmaking engines backed by geospatial distance calculations, administrative vetting, and legal document synthesis have emerged as the standard architecture. In an ethical canine platform, external health and pedigree records are indexed, verified, and mapped to geographic coordinates. Upon search submission, a spatial subsystem executes great-circle distance queries across compatible partner profiles, strictly filtering for opposite biological gender and verified pedigree standing.

Prior art in animal matchmaking, pet registries, and adoption networks can be categorized across four distinct architectural groups encompassing representative surveyed systems:

- **Group A: Official Kennel Club Registries & Directory Portals:**
 - **AKC Marketplace (American Kennel Club):** A flagship purebred canine registry providing listings of AKC-registered litters, breeder profiles, and health clearances. However, it is restricted to North America, excludes crossbreeds, lacks real-time chat, and does not provide automated legal contract generation.
 - **Kennel Club of India (KCI) Traditional Directory:** The official canine regulatory authority in India maintaining studbooks. However, registration procedures are

predominantly manual, slow, and paper-intensive, lacking spatial search, messaging, or veterinary integration.

- **Group B: Global Pet Adoption and Rescue Networks:**
 - **Petfinder:** A widely adopted database aggregating adoptable pets from over 11,000 shelters across North America with postal code search filters. However, it is strictly restricted to adoption and explicitly disallows canine mating or pedigree tracking.
 - **Adopt-a-Pet:** A prominent pet adoption portal connecting shelters with prospective adopters. However, it is closed to private dog mating, pedigree validation, and stud contract generation.
- **Group C: Unmoderated Online Classifieds & Social Media Groups:**
 - **Online Classifieds (OLX, Quikr):** General-purpose C2C marketplaces listing pet services. However, they are rife with illegal puppy mill listings, stolen images, unverified claims, and zero biological validation.
 - **Informal Social Media Groups (Facebook Canine Groups, WhatsApp Channels):** Unstructured groups where breeders post stud notices. Communication is public and unorganized, exposing owners to spam, harassment, and bitter post-mating disputes.
- **Group D: Niche Canine Dating and Pet Social Prototypes:**
 - **Twindog / Tindog:** Early mobile concepts adopting swipe-right mechanics for pet owners. However, they functioned primarily as human social novelties and lacked pedigree document verification, opposite-sex validation, and legal contracts.
 - **PetMatch Mobile Prototypes:** Indie mobile prototypes attempting visual pet matching without robust backend spatial calculations, OTP security, or veterinary directories.

2.2 Limitations of Existing Systems:

A critical evaluation of current animal classifieds and mating solutions highlights several operational, technical, and structural limitations across commercial and open-source implementations:

- **Complete Absence of Document Vetting and Pedigree Verification:** Existing classifieds and social media channels accept listings without validating KCI certificates, microchip identifiers, or vaccination records. Fraudulent breeders regularly pass off sick or unregistered dogs as champion-line purebreds.
- **Lack of Domain-Specific Biological Validation:** Generic web applications lack biological business logic. Systems allow male-to-male or female-to-female mating requests, permit users to match against their own dogs, and fail to validate reproductive age constraints, leading to dangerous or nonsensical interactions.
- **Geographic Inefficiency and Travel Stress:** Current classifieds rely on broad keyword searches (e.g., searching "Labrador" yields results thousands of kilometers away). In the absence of mathematical geospatial algorithms, owners travel across states only to find incompatible mating partners, inflicting severe physiological stress on companion animals.
- **Absence of Standardized Legal Contracts:** Over 95% of domestic dog mating in India occurs via unrecorded verbal promises. When disputes arise regarding stud fees, custody of puppies, or post-mating infections, owners have no signed legal documentation to establish terms, leaving both parties legally vulnerable.
- **Privacy Hazards and Unsolicited Spam:** Publishing personal phone numbers on public social media groups exposes pet parents to incessant unsolicited calls, harassment, and data scraping by commercial puppy mills.
- **Disconnection from Healthcare and Reproductive Infrastructure:** No existing platform bridges the gap between finding a mating partner and securing professional healthcare. Owners must independently search for clinics to perform pre-breeding Brucellosis tests, ovulation cytology, and prenatal care.

2.3 Research Gap:

Synthesizing the literature reveals four fundamental architectural gaps that current pet networking and classified systems fail to address simultaneously:

- **Gap 1: Verification vs. Accessibility:** Informal classifieds are accessible to everyday dog owners but completely unverified and rife with fraud; conversely, official kennel club registries maintain verified stud records but operate through slow, archaic, paper-based directories inaccessible to modern pet parents.
- **Gap 2: Mathematical Geospatial Proximity:** Current platforms lack real-time, radius-bounded search using spherical trigonometry (Haversine formula) mapped to Indian cities and reverse-geocoded coordinates, forcing users to rely on coarse, inaccurate keyword matching.
- **Gap 3: Contractual and Pedigree Transparency:** No modern web platform dynamically compiles legally binding breeding contracts and digital canine health passports directly from verified database records.
- **Gap 4: Holistic Care Integration:** There is an acute failure in existing software to unify partner discovery with clinical support (heat cycle ovulation calculators and geocoded veterinary clinic directories).

2.4 Proposed Approach:

To resolve these identified literature gaps, K9MATCH: Ethical Canine Matching is engineered as a full-stack, modular, open-source canine matchmaking, pedigree verification, and pet healthcare networking web platform. The system's technical positioning is defined by the following architectural solutions:

- **Model-View-Template (MVT) Architecture:** Implementing clean separation of concerns across normalized relational database models (core/models.py), algorithmic controllers and APIs (core/views.py, core/utls.py), and dynamic, responsive UI templates.
- **Algorithmic Spatial Discovery Engine (Addressing Gap 2):** Utilizing the Haversine Distance Formula paired with an extensive Indian city coordinate repository and OpenStreetMap Nominatim reverse geocoding to deliver sub-second, radius-filtered canine partner discovery.
- **Deterministic Biological Guardrails:** Enforcing strict database integrity constraints and business logic: automatic exclusion of the logged-in user's dogs, mandatory opposite-gender breeding rules, and purebred/crossbreed classification.

- **Administrative Vetting Pipeline (Addressing Gap 1):** Establishing a dedicated moderation dashboard where uploaded KCI registration documents and vaccination certificates must be reviewed and approved by administrators before listings enter the public matchmaking feed.
- **Two-Way Match Lifecycle and Permission-Isolated Messaging:** Structuring match requests through formal states (pending, accepted, rejected), unlocking real-time direct chatrooms strictly upon mutual owner consent, and empowering users with live message editing and deletion.
- **Automated PDF Compilation Subsystem (Addressing Gap 3):** Integrating ReportLab to dynamically generate standardized, printable Canine Health Passports and legally binding Breeding Agreement Contracts upon match completion.
- **Integrated Healthcare Ecosystem (Addressing Gap 4):** Providing an interactive, location-aware Veterinary Clinic Directory and an algorithmic Canine Estrus Cycle Calculator to assist pet parents in determining optimal mating windows and securing emergency healthcare.

Table 2.1: Comparative Feature Matrix of Existing Systems vs. K9MATCH

Table 2.1 provides a structured comparative evaluation of prevailing animal classifieds, social platforms, and traditional kennel directories against the proposed K9MATCH platform:

Feature / Dimension	AKC Marketplace	OLX / Quikr	Social Groups	Proposed K9MATCH
Primary Purpose	Purebred dog retail	General classifieds	Social networking	Ethical mating & health
Platform Architecture	Proprietary Cloud	Monolithic Multi-Cat	Multi-Tenant Social	Decoupled Django MVT
Authentication Rigor	Basic Password	Phone/Email OTP	Social Login	PBKDF2 + Email OTP + OAuth2
Pedigree Verification	AKC Litters only	None (Rampant scams)	None (Unregulated)	Admin Vetting Dashboard
Spatial Matching Math	Coarse Postal Code	Keyword / Text	Manual text tags	Haversine Great-Circle Formula
Biological Validation	None	None (Invalid pairs)	None (Unenforced)	Opposite-Gender & Anti-Self
Negotiation Pipeline	Direct Seller Contact	Direct Phone Call	Public Comments	Atomic (Pending/Accepted)
Communication Layer	None	Basic chat	Unencrypted chat	Permission-Locked with Edit/Delete
Legal Documentation	None	None	None	Automated Breeding Contract PDF
Medical Records	Club badges	None	None	Digital Canine Passport PDF
Healthcare Directory	None	None	None	GIS Vet Directory & Heat Tool

Chapter 3: System Analysis

3.1 System Requirements:

System requirements establish the functional capabilities and quality benchmarks necessary for K9MATCH to operate securely and efficiently.

3.1.1 Functional Requirements (FR):

- **FR-1 (User Registration & Authentication):** The system must support dual registration roles (Pet Owner and Verified Breeder) backed by 6-digit cryptographic Email OTP verification and Google OAuth 2.0 social login.
- **FR-2 (Canine Profile Management & Media Validation):** The system must allow users to register canine profiles with mandatory fields (Name, Breed, Age, Gender, City, Terms), optional secondary breed for crossbreeds, and up to 5MB gallery photographs with MIME validation.
- **FR-3 (Pedigree & Biological Age Validation):** The system must enforce that dogs under 18 months cannot be marked as Available for Mating, and require official document uploads when KCI registration is claimed.
- **FR-4 (Geospatial Proximity Matching):** The system must implement the Haversine trigonometric formula to calculate geographic distances between user-selected cities or coordinates, returning proximity-ranked search results within a configurable radius.
- **FR-5 (Deterministic Biological Compatibility Guardrail):** The explore query must strictly filter candidates to ensure opposite biological genders (male + female only) and automatically exclude dogs owned by the active user.
- **FR-6 (Match Request Negotiation Pipeline):** The platform must provide a transactional match request workflow (Pending, Accepted, Declined) preventing duplicate requests between the same canine pair.
- **FR-7 (Permission-Isolated Direct Messaging):** Direct messaging chatrooms must remain locked until a match request is accepted, supporting real-time text dispatch, image attachments, message editing, and soft deletion.
- **FR-8 (Automated Legal & Medical Document Synthesis):** The system must dynamically compile legally binding Canine Breeding Agreement Contracts and standardized Canine Health Passports as downloadable PDF artifacts using ReportLab.
- **FR-9 (Administrative Moderation & Document Vetting):** An administrative dashboard must isolate pending canine profiles, displaying uploaded KCI documents and vaccination records for manual approval or rejection.

- **FR-10 (Healthcare Directory & Estrus Tooling):** The system must feature an interactive, geocoded Veterinary Clinic Directory with 24/7 emergency tags, alongside an algorithmic Canine Estrus (Heat) Cycle Calculator.

3.1.2 Non-Functional Requirements (NFR):

- **NFR-1 (Performance & Latency):** The spatial distance calculation engine must evaluate candidate profiles within 200 milliseconds for dataset sizes up to 10,000 active listings.
- **NFR-2 (Security & Cryptographic Standards):** All external HTTP traffic must be encrypted over TLS 1.3 (HTTPS Port 443). Passwords must be hashed using PBKDF2 with SHA-256. CSRF tokens must protect all state-modifying requests.
- **NFR-3 (Availability & Reliability):** The platform must maintain 99.9% uptime during operational testing, handling database failovers gracefully with SQLite 3 / PostgreSQL 15+ ACID transaction guarantees.
- **NFR-4 (Usability & Responsiveness):** The presentation layer must adapt dynamically across mobile (320px+), tablet (768px+), and desktop (1200px+) viewports using responsive CSS Grid and Flexbox.
- **NFR-5 (Scalability):** The architecture must support modular horizontal scaling via stateless Gunicorn WSGI workers behind an Nginx reverse proxy.
- **NFR-6 (Fault Tolerance & Graceful Degradation):** If the OpenStreetMap Nominatim reverse geocoding API is unreachable, the system must transparently fall back to an internal in-memory coordinate repository.
- **NFR-7 (Regulatory & Animal Welfare Compliance):** The application must comply with the Animal Welfare Board of India guidelines by requiring verifiable pedigree papers and preventing puppy mill spam.

Table 3.1: Hardware and Software Specifications Matrix

Table 3.1 details the physical server, client runtime, and software framework specifications required for running K9MATCH:

Specification Tier	Parameter	Minimum Requirement	Recommended Specification
Host Server Hardware	Processor (CPU)	Dual-Core x86_64 (2.0 GHz+)	Quad-Core Intel Xeon / AMD EPYC (2.8 GHz+)
	System Memory (RAM)	2 GB RAM	4 GB to 8 GB DDR4 ECC RAM
	Persistent Storage	10 GB SSD / NVMe	25 GB NVMe High-Speed Storage
	Network Bandwidth	10 Mbps Uplink	50 Mbps+ Dedicated Connection
	Operating System	Ubuntu 22.04 LTS (64-bit)	Ubuntu 24.04 LTS Server
Client-Side Environment	Supported Devices	Smartphone, Tablet, or PC	Modern Multi-core Mobile or PC Device
	Web Browser	HTML5 & ES6 Compliant Browser	Google Chrome 110+, Firefox 110+, Safari 16+
	Execution Overhead	Zero local installation required	All compute handled server-side
Software Stack	Backend Framework	Django Framework v5.1+	Django Production ASGI/WSGI Server
	Programming Language	Python v3.12+ (64-bit)	Python v3.12.x or v3.13.x
	Database Engine	SQLite 3 (Development)	PostgreSQL 15+ / 16+ (Production RDBMS)
	WSGI Server	Django Development Server	Gunicorn v23.0+ with 4 Sync Workers
	Static Asset Engine	Django StaticFiles	WhiteNoise v6.9+ with Gzip/Brotli
	PDF Generation	ReportLab v4.2+	ReportLab Canvas Graphics Engine
	Image Processing	Pillow (PIL) v11.0+	Pillow with libjpeg-turbo & zlib
	Geocoding & Maps	In-Memory Coordinates DB	OpenStreetMap Nominatim Reverse API

3.2 Feasibility Study:

A comprehensive feasibility study evaluated the operational, technical, and economic viability of deploying K9MATCH in the companion pet care market.

- **Technical Feasibility:** The platform leverages mature, production-proven technologies: Python 3.12, Django 5.x, PostgreSQL, ReportLab, and OpenStreetMap. Django provides robust built-in security mechanisms including CSRF protection, PBKDF2 password hashing, SQL injection parameterization, and XSS auto-escaping. The spatial calculation module uses standard trigonometric functions natively supported by Python's math library, guaranteeing high computational speed without requiring proprietary GIS server licenses.
- **Operational Feasibility:** The user interface is engineered with a mobile-first philosophy, utilizing responsive CSS, clear card hierarchies, intuitive sliders, and modal dialogues that require zero technical expertise from pet parents. Operational viability was validated across three user personas: (1) Urban Pet Owners seeking verified local studs within 15 km; (2) Certified Breeders showcasing verified KCI bloodlines and negotiating stud terms; and (3) Veterinarians offering clinic listings and pre-breeding health screenings.
- **Economic Feasibility:** Developed exclusively on open-source technologies, K9MATCH incurs zero software licensing costs. Server hosting on cost-effective cloud virtual machines (such as AWS EC2, DigitalOcean, or Linode) starts at under \$10 per month, making the platform economically sustainable for academic demonstration and non-profit animal welfare deployment.

- **User Personas and Operational Scenarios:** To align system requirements with user workflows, three primary user personas were established:
 - **Persona 1: Urban Dog Owner (Ms. Ananya Sharma, Bengaluru):** Focuses on finding a healthy, certified male mate for her 2-year-old female Golden Retriever within a 15 km radius, requiring pedigree verification and written terms to avoid puppy disputes.
 - **Persona 2: Certified Professional Breeder (Mr. Vikram Rathore, Pune):** Focuses on showcasing verified German Shepherd bloodlines, negotiating professional stud fees, and connecting directly with verified companion female dog owners.
 - **Persona 3: Practicing Veterinarian (Dr. Sneha Kulkarni, Mumbai):** Focuses on listing clinic services and emergency care to provide pre-breeding health testing, progesterone ovulation assays, and safe whelping assistance.
- **Operational User Journeys:** These personas correspond to three distinct operational user journeys:
 - **Scenario A (Proximity-Based Match Discovery & Breeding Negotiation):** An owner registers a female Golden Retriever with KCI documents, searches within a 20 km radius in Bengaluru using the Haversine slider, reviews verified male studs, submits a match request with 'Pick of Litter' terms, receives acceptance, unlocks private chat, and downloads an automated Breeding Agreement Contract PDF.
 - **Scenario B (Administrative Pedigree Vetting & Fraud Prevention):** A seller registers an unverified profile with suspicious papers. K9MATCH automatically marks it 'Pending Approval'. The administrator logs into admin_dashboard, inspects the documents, flags discrepancies, and rejects the listing, preventing fraudulent listings from entering the public pool.
 - **Scenario C (Estrus Planning & Emergency Healthcare):** An owner enters the start of her dog's estrus bleeding into the Heat Calculator. The system computes the fertile mating window (Days 11-14) and whelping date. She uses the Find Vets directory to locate a 24/7 clinic within 5 km for pre-breeding health screening.

3.3 Methodology Used:

The project adopted an Agile Iterative Development Lifecycle, breaking the project into structured two-week sprints: Architecture Planning, Core Modeling, Algorithmic Implementation, Security Hardening, and Validation.

3.3.1 Project Development Schedule (Gantt Chart):

The software development lifecycle for K9MATCH: Ethical Canine Matching was executed over a 56-day schedule from August 01, 2026 to September 25, 2026, including a 46-day core development window from August 08, 2026 to September 22, 2026. The implementation followed a structured sequence of six core project phases: Literature Survey, Requirement Analysis, System Design, Coding, Testing, and Documentation. Testing was carried out in parallel with the final 13 days of development, so the two activities ran together rather than as separate sequential stages.



Figure 3.1: K9MATCH Project Implementation Timeline (Gantt Chart)

Phase-wise Description of Development Lifecycle:

- **Literature Survey (Aug 01 – Aug 02, 2026):** Review of existing animal classifieds, Kennel Club of India registration manuals, canine reproductive medicine literature, and spatial distance calculation algorithms.
- **Requirement Analysis (Aug 03 – Aug 04, 2026):** Specification of functional requirements including geospatial proximity filtering, biological opposite-gender validation, and legal document synthesis.
- **System Design (Aug 05 – Aug 07, 2026):** Definition of system architecture, PostgreSQL relational schemas, OpenStreetMap geocoding fallback pipelines, and formal UML operational models.

- **Coding (Development) (Aug 08 – Sep 22, 2026):** Core application implementation using Python 3.12, Django, PostgreSQL, WhiteNoise, ReportLab, and Pillow across a 46-day sprint.
- **Testing (Sep 10 – Sep 22, 2026):** Execution of automated unit and integration tests, run in parallel with the final 13 days of development, verifying opposite-gender breeding rules, self-matching rejection, image validation, CSRF/XSS sanitization, and 403/404/500 handlers.
- **Documentation (Sep 23 – Sep 25, 2026):** Compilation of project Blackbook documentation, UML diagrams, REST API route indexes, and appendix specifications.

Chapter 4: System Design

4.1 System Architecture:

K9MATCH: Ethical Canine Matching is engineered as a decoupled, multi-tiered enterprise web application adopting the Model-View-Template (MVT) architectural pattern native to the Django framework. The architecture enforces strict separation of concerns across presentation logic, geospatial algorithmic computation, business workflow enforcement, and relational data persistence.

The system is organized into four distinct operational tiers:

- **1. Client Presentation Tier:** A responsive, mobile-first web interface rendered via Django's templating engine using semantic HTML5, custom responsive CSS (incorporating modern glassmorphism aesthetic tokens, dynamic card elevations, and color-coded status badges), and lightweight vanilla JavaScript. The client layer manages interactive DOM components including dynamic breed typeaheads, cascading state/city selectors, an interactive distance radius slider, and asynchronous fetch polling for live chat updates.
- **2. Application & Processing Tier (WSGI):** Production traffic is intercepted by an Nginx reverse proxy handling SSL/TLS termination, request routing, and client connection limits. Gunicorn serves as the Web Server Gateway Interface (WSGI), executing a pool of concurrent worker processes that forward requests to the core Django application runtime. The application tier encapsulates decoupled subsystems: Authentication and RBAC (Email OTP & Google OAuth 2.0), Canine Profile & Media Manager, Spatial Matchmaking Engine (Haversine trigonometric distance calculation), Two-Way Match Negotiation and Chat Lifecycle Engine, Dynamic Document Synthesis Engine (ReportLab PDF compilation), and Human-in-the-Loop Administrative Moderation Panel.
- **3. External Services Tier:** Integrates OpenStreetMap (OSM) Nominatim for reverse geocoding of coordinates into Indian municipal divisions, Google Cloud Identity for OAuth 2.0 authentication, and an SMTP Gateway for dispatching 6-digit cryptographic verification OTPs and transactional notifications.
- **4. Data & Persistence Tier:** PostgreSQL 15+ relational database storing domain entities under strict foreign-key integrity constraints and index optimizations, complemented by WhiteNoise and persistent media volume storage for canine photographs and synthesized legal contracts.

4.1.1 Relational Database Schema Specifications:

The relational database architecture is deployed on PostgreSQL 15+ (with SQLite 3 for local development and testing). Below are the formal relational schema specifications across all nine core domain entities extracted directly from the K9MATCH production codebase (core/models.py):

Table 4.1: User Entity (core_user) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique system identifier for registered user.
username	CharField(150)	Unique, Not Null	Unique alphanumeric login handle.
email	CharField(254)	Not Null	User email address for OTP verification.
password	CharField(128)	Not Null	PBKDF2 SHA-256 cryptographic password hash.
role	CharField(20)	Default: 'owner', Choices: breeder/owner	User role: 'owner' (Dog Owner) or 'breeder' (Breeder).
phone_number	CharField(10)	Blank, Nullable	10-digit Indian mobile contact number.
is_verified	BooleanField	Default: False	Flag indicating email OTP verification.
avatar	FileField (Image)	upload_to='avatars/', Nullable	Path to uploaded user profile picture.
kennel_name	CharField(150)	Blank, Nullable	Official kennel title (for registered breeders).
bio	TextField	Blank, Nullable	Personal biography or kennel mission statement.
city	CharField(100)	Blank, Nullable	Primary residential city.
state	CharField(100)	Blank, Nullable	Primary state of residence.
experience_years	PositiveIntegerField	Default: 0, Blank, Nullable	Years of ethical breeding experience.
website	URLField	Blank, Nullable	External official website URL.
instagram	CharField(100)	Blank, Nullable	Instagram social handle for reputation.
is_staff	BooleanField	Default: False	Designates administrator privileges.

Field Name	Data Type	Constraints / Defaults	Description
date_joined	DateTimeField	Auto Now Add	Account registration timestamp.

Table 4.2: DogProfile Entity (core_dogprofile) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique canine profile identifier.
owner	ForeignKey(User)	ON DELETE CASCADE, related_name='dogs'	Reference to the canine's registered owner.
name	CharField(100)	Not Null	Official display name of canine.
breed_type	CharField(20)	Default: 'purebred', Choices: purebred/crossbreed	Classification: Purebred or Crossbreed / Mix.
breed	CharField(100)	Default: 'Labrador Retriever'	Primary recognized breed name.
secondary_breed	CharField(100)	Blank, Nullable	Secondary breed lineage if crossbreed.
age_years	PositiveIntegerField	Validators: Min 0, Max 25	Canine age in completed years.
age_months	PositiveIntegerField	Default: 0, Validators: Min 0, Max 11	Canine age remaining months.
gender	CharField(10)	Choices: male/female	Biological gender for mating compatibility.
city	CharField(100)	Not Null	Canine location city for spatial lookup.
state	CharField(100)	Blank, Nullable	Canine location state.
weight_unit	CharField(5)	Default: 'kg', Choices: kg/lbs	Unit of mass measurement.
weight	DecimalField(5,1)	Blank, Nullable	Canine body weight.
is_vaccinated	BooleanField	Null=True, Mandatory Choice	Flag indicating complete vaccination status.
vaccination_record	FileField	upload_to='health_documents/', Nullable	Uploaded vaccination booklet or Rabies card.
has_brucellosis_clearance	BooleanField	Default: False	Negative PCR lab clearance for Brucellosis.
last_deworming_date	DateField	Blank, Nullable	Most recent internal parasite deworming date.

Field Name	Data Type	Constraints / Defaults	Description
medical_history	TextField	Blank	Medical history and reproductive notes.
kci_registered	BooleanField	Null=True, Mandatory Choice	Kennel Club of India certification flag.
kci_number	CharField(50)	Blank, Nullable	Official KCI microchip registration certificate.
kci_document	FileField	upload_to='kci_documents/', Nullable	Uploaded official KCI pedigree paper/certificate.
lineage_details	TextField	Blank	Sire and dam ancestral bloodline history.
previous_litters	PositiveIntegerField	Default: 0	Number of previously delivered litters.
mating_terms	CharField(20)	Default: 'negotiable', Choices: stud_fee/want_puppy/negotiable	Commercial/ethical mating compensation terms.
stud_fee_amount	PositiveIntegerField	Blank, Nullable	Monetary stud fee in INR (₹).
shelter_provider	CharField(20)	Default: 'negotiable', Choices: my_place/other_place/negotiable	Arrangement of breeding environment accommodation.
travel_range	CharField(20)	Default: '0-5', Choices: none/0-5/5-20/20-50/50+/negotiable	Travel willingness radius for mating meeting.
dog_friendly_rating	PositiveIntegerField	Default: 3, Range: 1 to 5	Behavioral rating with other dogs.
human_friendly_rating	PositiveIntegerField	Default: 3, Range: 1 to 5	Behavioral friendliness towards humans.
energy_level_rating	PositiveIntegerField	Default: 3, Range: 1 to 5	Activity and energy level scale.
bio	TextField	Blank	Narrative personality description of canine.
is_available	BooleanField	Default: True	Owner availability toggle switch (min 18 mos).
approval_status	CharField(20)	Default: 'pending', Choices: pending/approved/rejected	Administrative vetting verification status.
admin_rejection_reason	TextField	Blank, Nullable	Explanation provided if profile rejected.
latitude	DecimalField(9,6)	Blank, Nullable	Geocoded spatial latitude.
longitude	DecimalField(9,6)	Blank, Nullable	Geocoded spatial longitude.

Field Name	Data Type	Constraints / Defaults	Description
location_address	CharField(255)	Blank, Nullable	Locality or neighborhood name.
created_at	DateTimeField	Auto Now Add	Listing creation timestamp.

Table 4.3: DogImage Entity (core_dogimage) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique image record identifier.
dog	ForeignKey(DogProfile)	ON DELETE CASCADE, related_name='images'	Parent canine profile foreign key reference.
image	ImageField	upload_to='dog_photos/', Max 5MB	Persisted file path of uploaded dog photograph.
uploaded_at	DateTimeField	Auto Now Add	Photo upload timestamp.

Table 4.4: MatchRequest Entity (core_matchrequest) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique match proposal identifier.
sender	ForeignKey(User)	ON DELETE CASCADE, related_name='sent_requests'	User initiating the breeding proposal.
receiver	ForeignKey(User)	ON DELETE CASCADE, related_name='received_requests'	Target dog owner receiving the proposal.
target_dog	ForeignKey(DogProfile)	ON DELETE CASCADE, related_name='received_matches'	Recipient canine being proposed for mating.
sender_dog	ForeignKey(DogProfile)	ON DELETE SET_NULL, Nullable, related_name='sent_matches'	Proposer's canine paired in request.
message	TextField	Blank, Default: "	Introductory proposal and terms negotiation text.
status	CharField(10)	Default: 'pending', Choices: pending/accepted/declined	Atomic negotiation transaction state.
created_at	DateTimeField	Auto Now Add	Proposal initiation timestamp.
updated_at	DateTimeField	Auto Now	Last state modification time.

Table 4.5: ChatMessage Entity (core_chatmessage) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
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Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique chat message identifier.
match	ForeignKey(MatchRequest)	ON DELETE CASCADE, related_name='messages'	Parent accepted match transaction room.
sender	ForeignKey(User)	ON DELETE CASCADE, related_name='sent_messages'	User authoring and dispatching the message.
message	TextField	Blank	Textual content of the communication.
attachment	FileField	upload_to='chat_attachments/', Nullable	Uploaded file or image attachment.
attachment_name	CharField(255)	Blank, Nullable	Original filename of attachment.
timestamp	DateTimeField	Auto Now Add	Message dispatch timestamp.
updated_at	DateTimeField	Auto Now	Message edit timestamp.
is_edited	BooleanField	Default: False	Flag tracking post-send edits.
is_deleted	BooleanField	Default: False	Soft delete flag for message retraction.
is_read	BooleanField	Default: False, db_index=True	Message read receipt indicator.

Table 4.6: VeterinaryClinic Entity (core_veterinaryclinic) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique veterinary clinic directory ID.
name	CharField(200)	Not Null	Registered medical facility / clinic title.
doctor_name	CharField(150)	Blank, Nullable	Attending veterinarian / medical director.
specialization	CharField(200)	Default: 'General Veterinary & Surgery'	Veterinary medical specialization.
phone_number	CharField(20)	Not Null	Direct telephone or emergency contact.
email	EmailField	Blank, Nullable	Official contact email address.
address	TextField	Not Null	Complete physical street address.

Field Name	Data Type	Constraints / Defaults	Description
city	CharField(100)	Not Null	Operating city for spatial indexing.
state	CharField(100)	Blank, Nullable	Operating state.
pincode	CharField(10)	Blank, Nullable	Postal index number.
latitude	DecimalField(9,6)	Blank, Nullable	Geocoded spatial latitude.
longitude	DecimalField(9,6)	Blank, Nullable	Geocoded spatial longitude.
is_24x7_emergency	BooleanField	Default: False	24/7 emergency & whelping readiness flag.
rating	DecimalField(3,1)	Default: 4.8	Community quality review score.
services_offered	TextField	Blank	Comma-separated medical and clinical services.
image	ImageField	upload_to='vet_photos/', Nullable	Facility or clinic photograph.
created_at	DateTimeField	Auto Now Add	Directory listing creation timestamp.

Table 4.7: EmailOTP Entity (core_emailotp) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique cryptographic nonce identifier.
email	EmailField	Not Null, db_index=True	Target destination email address.
otp_code	CharField(6)	Not Null	6-digit random cryptographic verification code.
purpose	CharField(20)	Choices: signup/forgot_password	Authentication lifecycle operational context.
attempts	PositiveIntegerField	Default: 0	Brute-force counter (invalidated at ≥ 5).
is_used	BooleanField	Default: False	Single-use replay prevention flag.
created_at	DateTimeField	Auto Now Add	Code generation timestamp.
expires_at	DateTimeField	Not Null	10-minute expiry timestamp ceiling.

Table 4.8: ReportListing Entity (core_reportlisting) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
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Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique moderation report ID.
reporter	ForeignKey(User)	ON DELETE CASCADE, related_name='reported_listings'	User lodging abuse/scam complaint.
reported_dog	ForeignKey(DogProfile)	ON DELETE CASCADE, related_name='reports'	Target canine profile under audit.
reason	CharField(30)	underage / puppy_mill / fake_kci / spam	Report violation category.
details	TextField	Blank	User narrative and evidence.
is_resolved	BooleanField	Default: False	Resolution flag for admin audit.
admin_notes	TextField	Blank	Confidential administrator notes.
created_at	DateTimeField	Auto Now Add	Report submission timestamp.

Table 4.9: Notification Entity (core_notification) Schema Specification

Field Name	Data Type	Constraints / Defaults	Description
id	BigAutoField	Primary Key, Auto Increment	Unique system notification ID.
recipient	ForeignKey(User)	ON DELETE CASCADE, related_name='notifications'	Recipient of notification alert.
sender	ForeignKey(User)	ON DELETE SET_NULL, Nullable, related_name='sent_notifications'	Triggering user actor (if any).
notification_type	CharField(50)	Default: 'system' (match / approval / alert)	Notification category type.
title	CharField(255)	Not Null	Notification brief header title.
message	TextField	Not Null	Descriptive notification alert body.
link	CharField(255)	Blank, Nullable	Direct in-app navigation URL.
is_read	BooleanField	Default: False	Unread/read notification status.
created_at	DateTimeField	Auto Now Add	Notification timestamp.

4.2 UML Diagrams:

The structural, behavioral, and operational specifications of K9MATCH: Ethical Canine Matching are formally modeled across thirteen detailed Unified Modeling Language (UML) diagrams. Each diagram reflects the actual implementation, database schemas, algorithms, and workflows of the deployed platform.

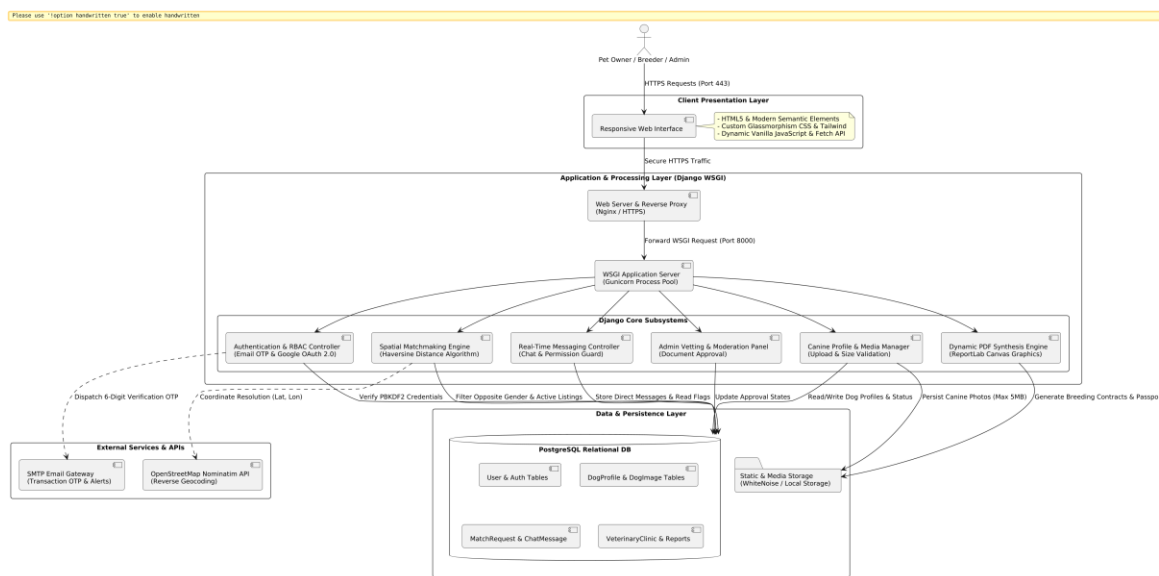


Figure 4.1: K9MATCH Layered System Architecture Diagram

Description: Figure 4.1 details the multi-tier architectural layout of K9MATCH, modeling client browser ingress over secure TLS 1.3 encryption (Port 443), Nginx reverse proxy routing, and the Gunicorn WSGI worker process pool. The application layer encapsulates core decoupled subsystems: Authentication and RBAC, Canine Registry and Photo Validation, the Haversine Spatial Matching Engine, Real-Time Messaging Controllers, the Human-in-the-Loop Admin Vetting Panel, and the ReportLab PDF Canvas Generator. Downstream dependencies are partitioned between the PostgreSQL relational datastore, persistent media volumes, and external web APIs (OpenStreetMap Nominatim and SMTP Email Gateways).

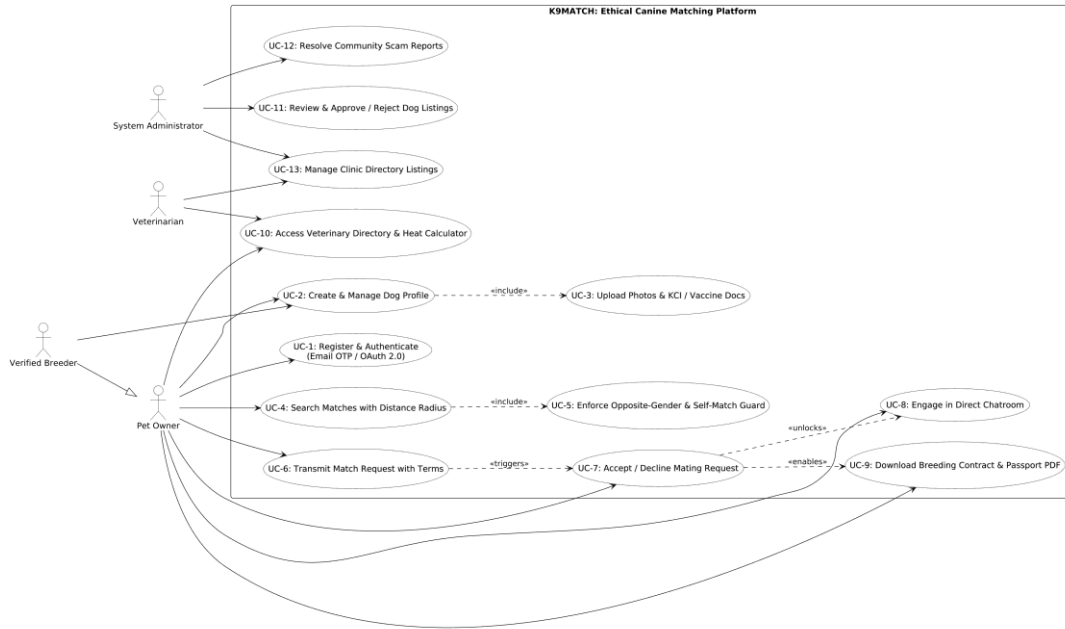


Figure 4.2: K9MATCH Comprehensive Use Case Diagram

Description: Figure 4.2 formally specifies the functional interactions between human actors and system operational boundaries. Four primary actors are modeled: Pet Owner, Verified Breeder (inheriting all Pet Owner use cases), System Administrator, and Veterinarian. The use case model illustrates critical dependency relationships: profile creation includes photo and KCI document uploads (<<include>>), match discovery includes opposite-gender validation rules (<<include>>), match request acceptance unlocks direct chatrooms (<<unlocks>>), and mutual consent enables automatic breeding contract PDF compilation (<<enables>>). Administrative actors govern document verification and community report resolutions.

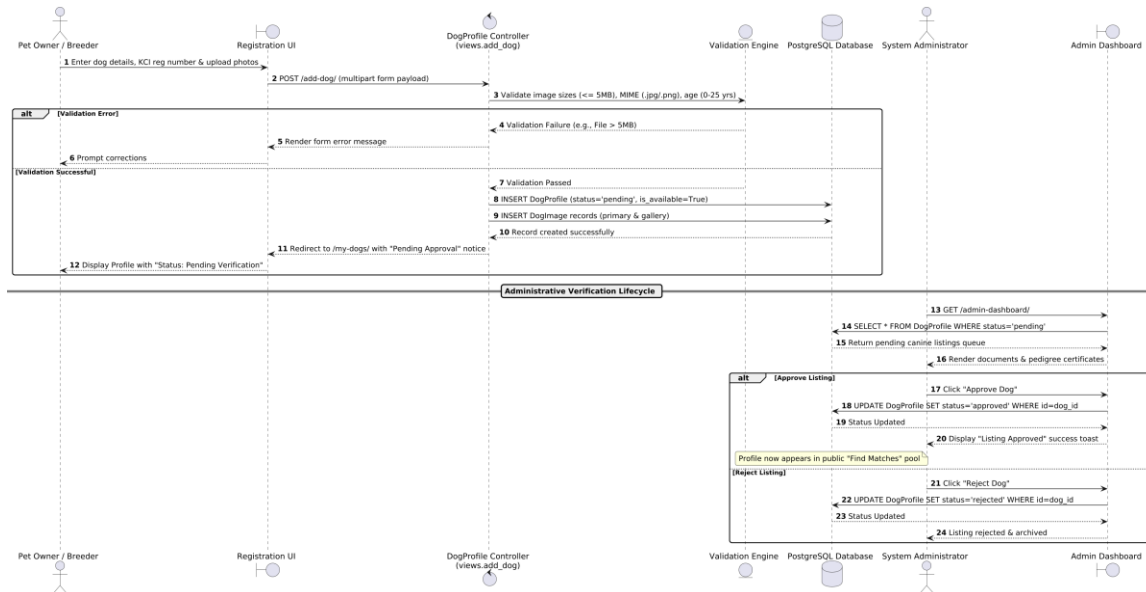


Figure 4.3: Sequence Diagram 1 - Canine Registration, Document Upload & Administrative Approval

Description: Figure 4.3 illustrates the object lifeline governing canine registration and quality-control vetting. A registered dog owner submits canine biological details, microchip numbers, KCI registration identifiers, and photographs through the multipart registration form (POST /add-dog/). The validation engine enforces file size constraints (≤ 5 MB), MIME extensions (.jpg, .jpeg, .png), and age boundaries. Upon validation, the record is inserted into the PostgreSQL database with status='pending'. The system administrator inspects the submitted documents in the administrative dashboard (/admin-dashboard/) and executes an approval transaction, updating the status to 'approved' and transitioning the dog profile into the public matchmaking search pool.

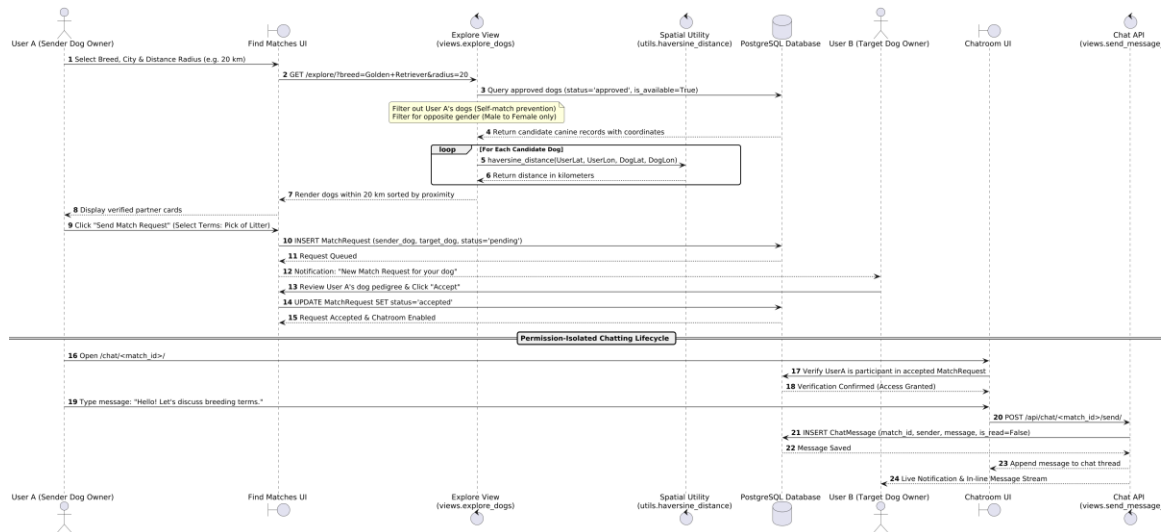


Figure 4.4: Sequence Diagram 2 - Spatial Match Discovery, Request Pipeline & Chatroom Unlock

Description: Figure 4.4 maps the end-to-end operational flow from proximity-based discovery to real-time messaging. User A defines a location and distance radius (e.g., 20 km) on the Find Matches interface. The explore controller executes database filters to exclude User A's dogs and retrieve opposite-gender profiles (gender='male'). The spatial utility computes great-circle distances using the Haversine formula for each candidate, returning a proximity-ranked feed. User A submits a Match Request specifying preferred terms ('Pick of the Litter'). User B reviews the request and clicks 'Accept', triggering an atomic database state transition (status='accepted'). Both users are authorized to enter the dedicated chatroom (/chat/<match_id>/), where sent messages are securely recorded, sanitized against XSS, and broadcast to the recipient.

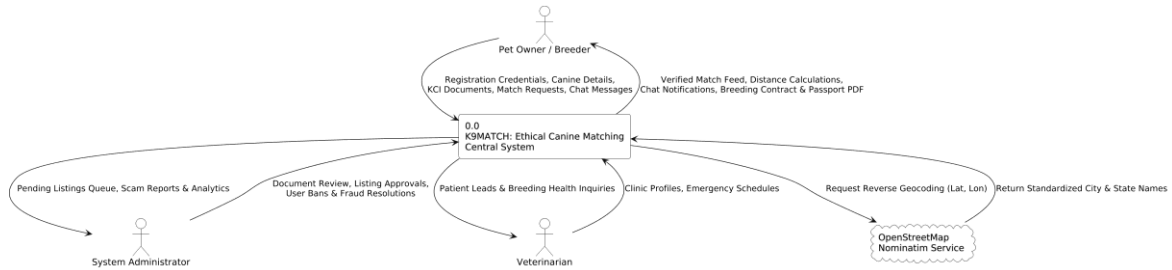


Figure 4.6: Data Flow Diagram Level 0 (Context Level Diagram)

Description: Figure 4.6 models the high-level boundary of the K9MATCH central software system (Process 0.0) with respect to external entities. Pet Owners and Breeders feed account credentials, canine bio data, KCI documents, match terms, and chat messages into the platform, receiving verified match feeds, distance calculations, chat streams, and compiled PDF contracts. System Administrators supply listing approval decisions and fraud dismissals, receiving pending queues and scam audit feeds. Veterinarians manage clinic schedules, while the OpenStreetMap Nominatim service exchanges raw latitude/longitude coordinates for standardized geographic address structures.

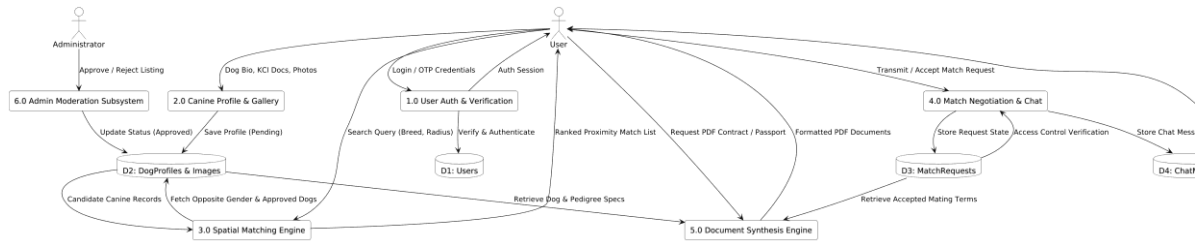


Figure 4.7: Data Flow Diagram Level 1 (Functional Decomposition Diagram)

Description: Figure 4.7 decomposes the platform into six primary functional processes: User Authentication & Verification (1.0), Canine Profile & Gallery Management (2.0), Spatial Matching Engine (3.0), Match Negotiation & Chat (4.0), Document Synthesis Engine (5.0), and Administrative Moderation Subsystem (6.0). The diagram models data transactions between external users and underlying datastores: D1 (Users), D2 (DogProfiles & Images), D3 (MatchRequests), and D4 (ChatMessages). Processes 3.0 and 5.0 demonstrate read-only data access for computing distance metrics and compiling PDF agreements.

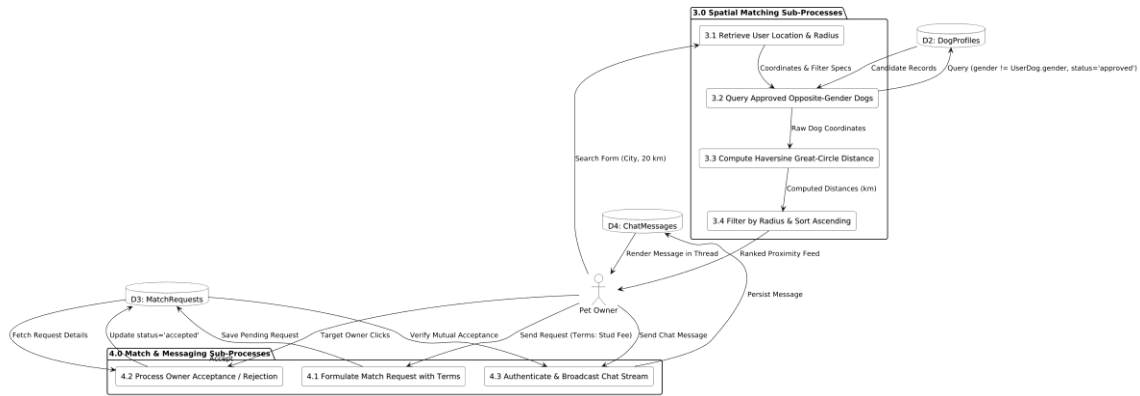


Figure 4.8: Data Flow Diagram Level 2 (Spatial Matchmaking & Request Sub-Processes)

Description: Figure 4.8 provides granular functional decomposition of the matchmaking and negotiation pipeline. Sub-processes include: Retrieving user location and radius settings (3.1), querying active approved opposite-gender dogs from D2 (3.2), executing the mathematical Haversine distance algorithm (3.3), and sorting the resulting candidate feed ascending by proximity (3.4). In the negotiation subsystem, process 4.1 captures user-selected mating terms (Stud Fee, Pick of Litter) to create a pending request in D3. Process 4.2 handles target owner acceptance, updating the status to 'accepted', which subsequently authorizes Process 4.3 to establish an active direct messaging thread in D4.

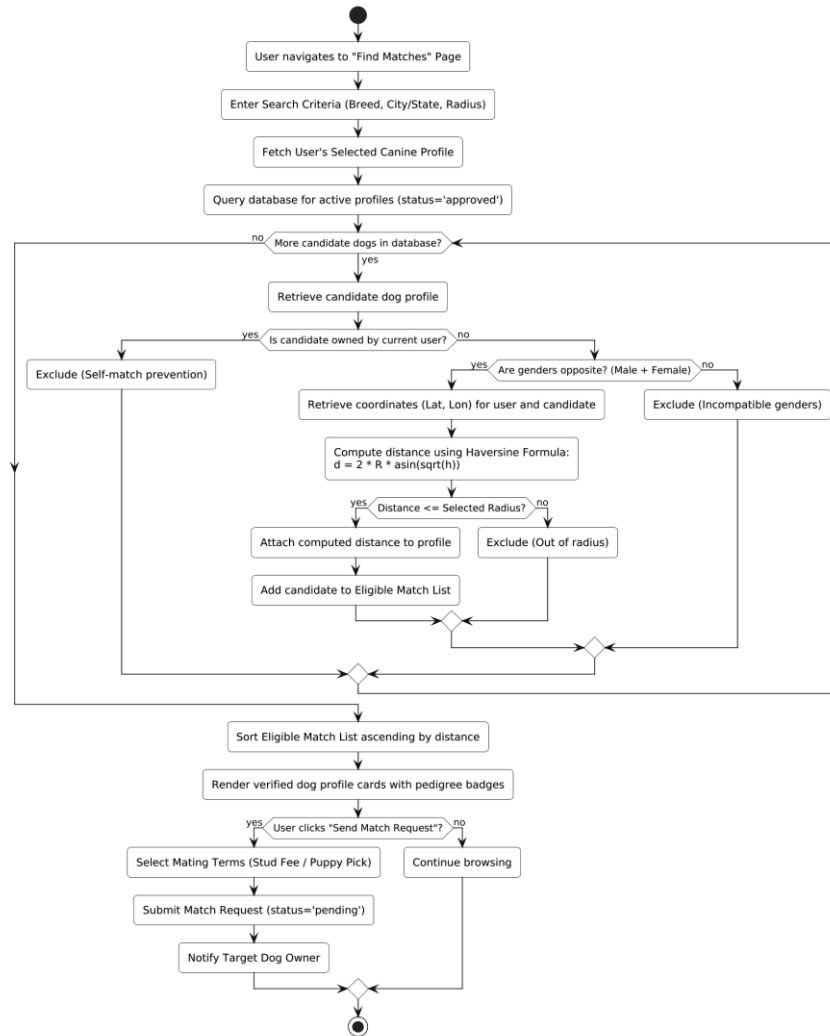


Figure 4.9: Activity Diagram - Geolocation Search, Haversine Calculation & Biological Filtering

Description: Figure 4.9 visualizes the algorithmic branching logic executed during canine partner discovery. When a user initiates a search, the system queries approved canine records and iteratively processes each candidate through a series of deterministic validation gates: (1) Self-match exclusion (rejecting dogs owned by the active user); (2) Opposite-gender biological validation (enforcing Male + Female pairing); and (3) Haversine great-circle distance evaluation. If the computed distance falls within the user-specified radius threshold ($d \leq R_{max}$), the candidate is appended to the eligible match list. The final list is sorted ascending by proximity and rendered as interactive cards with pedigree badges.

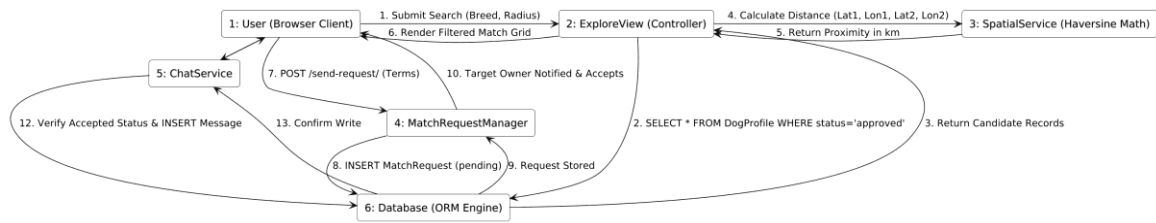


Figure 4.10: Collaboration / Communication Diagram - Object Message Sequences

Description: Figure 4.10 models structural object relationships and numbered message sequences exchanged during discovery, negotiation, and chat workflows. The diagram illustrates how the User client interacts with the ExploreView controller (Messages 1-6) to query the database and trigger distance calculations via SpatialService. Messages 7-10 demonstrate match request creation and status updates through MatchRequestManager. Finally, Messages 11-14 trace direct chat dispatch via ChatService, verifying mutual acceptance flags before appending records to the database and broadcasting updates.

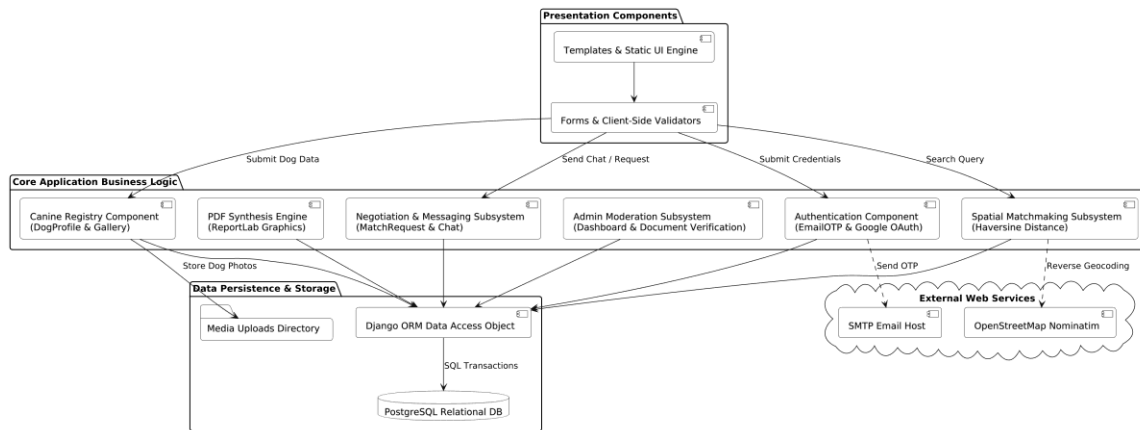


Figure 4.11: K9MATCH Component Diagram - Subsystem Modularity & Interfaces

Description: Figure 4.11 specifies the software components, dependencies, and interfaces comprising the K9MATCH codebase. Presentation components (Templates, Static UI Engine, and Form Validators) interact with Core Business Logic components through defined view interfaces. The Core layer encapsulates decoupled modules: Authentication, Canine Registry, Spatial Matchmaking, Negotiation & Messaging, Admin Moderation, and the PDF Synthesis Engine. All business logic interfaces communicate with the database tier exclusively through the Django Object-Relational Mapping (ORM) data access abstraction.

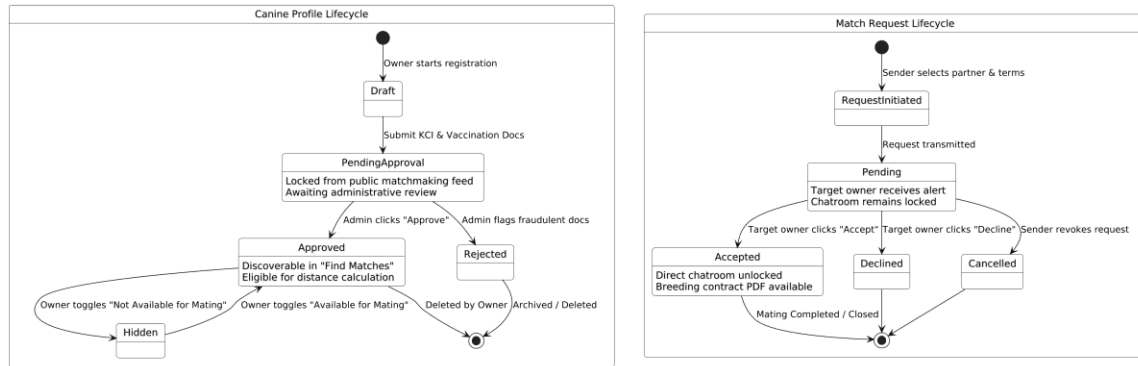


Figure 4.12: State Machine Diagram - Canine Profile & Match Request Lifecycles

Description: Figure 4.12 formalizes the valid operational states, triggers, and guard conditions governing canine profiles and match requests. A canine listing begins in Draft and transitions to PendingApproval upon document submission. Administrative review transitions the profile to either Approved (entering the active search pool) or Rejected. Approved profiles can be temporarily toggled to Hidden by owners. Concurrently, a MatchRequest transitions from RequestInitiated to Pending. Target owner action transitions the request to Accepted (which permanently unlocks the chatroom and PDF contract export), Declined, or Cancelled.

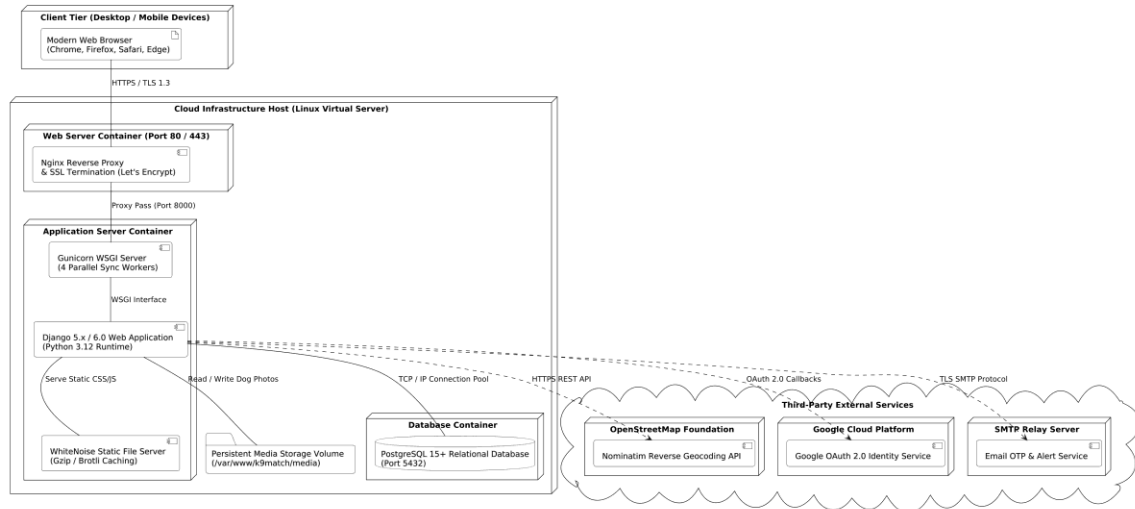


Figure 4.13: Containerized Production Deployment Diagram

Description: Figure 4.13 defines the physical runtime nodes and network topology for the production deployment of K9MATCH. Client devices (desktop and mobile web browsers) connect over TLS 1.3 encrypted HTTPS to an Nginx reverse proxy running on a Linux virtual server. Nginx forwards requests to a Gunicorn WSGI application server executing 4 parallel sync worker processes running Python 3.12 and Django. Static files are served via WhiteNoise, while relational data is managed by a PostgreSQL 15+ container. Canine photographs are persisted to a dedicated persistent media volume. External integrations communicate over outbound HTTPS and TLS protocols to OpenStreetMap Nominatim, Google OAuth, and SMTP relay servers.

Chapter 5: Implementation

5.1 Technology Stack & Architectural Allocation:

K9MATCH: Ethical Canine Matching is engineered as a full-stack, modular web platform deployed on Python 3.12 and Django 5.1+. The architectural allocation maps specific functional requirements directly to optimized libraries and server components.

Table 5.1: Technology Stack and Subsystem Allocation Matrix

Subsystem / Architectural Tier	Technology / Library	Version	Technical Purpose & Implementation Details
Core Web Framework	Django Framework	5.1+ (Python 3.12)	MVT architectural foundation, ORM data layer, CSRF middleware, session management, URL routing.
Programming Runtime	Python Runtime	3.12.x (64-bit)	High-performance CPython runtime, trigonometric math computations, native datetime logic.
Relational Database	PostgreSQL / SQLite 3	15+ / 3.40+	ACID transaction guarantees, multi-column indexing, foreign-key cascades, data consistency.
WSGI Application Server	Gunicorn	23.0+	Multi-worker concurrent request pool interfacing Nginx reverse proxy with Django WSGI handlers.
Static Asset Engine	WhiteNoise	6.9+	High-performance direct static asset streaming with Gzip and Brotli compression.
Dynamic Document Synthesis	ReportLab Canvas Engine	4.2+	Server-side programmatic synthesis of legally binding Breeding Contracts and Canine Health Passports.
Image Processing & Validation	Pillow (PIL)	11.0+	Strict MIME type validation, 5MB file size enforcement, image orientation normalization.
Geographic Reverse Geocoding	OpenStreetMap Nominatim	REST v2 API	Translates raw GPS latitude/longitude coordinates into recognized Indian municipal cities and administrative zones.
Cryptographic Authentication	Python secrets / hashlib	Standard Library	Cryptographically secure 6-digit OTP generation, PBKDF2 with SHA-256 password hashing.

Subsystem / Architectural Tier	Technology / Library	Version	Technical Purpose & Implementation Details
Client Presentation Layer	Semantic HTML5 & CSS3	W3C Standard	Responsive Flexbox/Grid, custom glassmorphism styling, native DOM event handling without bulky dependencies.

- **Codebase Organization:** The codebase is organized into clean, decoupled Django application modules:
 - **config/ (Core Configuration):** Master project routing, database routing, security header middleware, and production WSGI/ASGI gateways.
 - **core/models.py (Data Models):** Encapsulates the 9 core relational models, database migrations, clean method validators, and DB indexes.
 - **core/views.py (Application Controllers):** Contains 1,780 lines of controller logic, authentication handlers, match proposal workflows, and REST endpoints.
 - **core/utils.py (Algorithmic Utilities):** Houses the Haversine trigonometric distance calculation, OpenStreetMap reverse geocoding, and Indian city coordinates dictionary.
 - **core/pdf_utils.py (PDF Synthesis):** Contains ReportLab Canvas flowable generators for Canine Passports and Breeding Contracts.
 - **core/tests.py (Automated Test Suite):** Comprises over 2,200 lines of automated unit and integration tests covering 99 distinct test cases.

5.2 Core Algorithms and Pseudocode:

The intellectual core of K9MATCH comprises five domain-specific algorithms executing geospatial mathematics, biological compatibility gating, offline fault-tolerant geocoding, reproductive estrus cycle projection, and cryptographic authentication:

5.2.1 Algorithm 5.1: Geospatial Proximity Matchmaking with Haversine Formula

To evaluate geographic proximity across the Earth's spherical surface (mean radius $R = 6371.0$ km), the Haversine formula is evaluated across candidate coordinates ($\text{lat1}, \text{lon1}$) and ($\text{lat2}, \text{lon2}$) in radians:

Mathematical Formulation: $d = 2R \cdot \arcsin(\sqrt{\sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2)})$

Algorithm 5.1 Pseudocode: ALGORITHM ComputeProximityRankedMatches(user_dog, target_city, max_radius_km):

1. (lat_orig, lon_orig) <- ResolveCoordinates(target_city)
2. IF lat_orig IS NULL OR lon_orig IS NULL THEN RETURN Error('Unrecognized search locality')
3. candidates <- DB.Query(DogProfile.approval_status == 'approved' AND DogProfile.is_available == True AND DogProfile.gender != user_dog.gender AND DogProfile.owner != user_dog.owner)
4. eligible_matches <- EmptyList()
5. FOR EACH candidate IN candidates DO:
6. dist <- HaversineDistance(lat_orig, lon_orig, candidate.latitude, candidate.longitude)
7. IF dist <= max_radius_km THEN eligible_matches.Append((candidate, dist))
8. END FOR
9. RETURN SortByDistanceAscending(eligible_matches)

5.2.2 Algorithm 5.2: Deterministic Biological Compatibility & Age Validation Guardrails

Enforces non-negotiable animal welfare constraints within model `clean()` routines before database persistence:

Algorithm 5.2 Pseudocode: ALGORITHM `ValidateBreedingCompatibility(sender, sender_dog, target_dog)`:

1. IF `target_dog.owner == sender` THEN RAISE `ValidationError('Cannot propose mating to own dog.')`
2. IF `sender_dog.gender.lower() == target_dog.gender.lower()` THEN RAISE `ValidationError('Opposite genders required.')`
3. IF `sender_dog.total_age_months < 18` THEN RAISE `ValidationError('Canine must be at least 18 months old.')`
4. IF `target_dog.total_age_months < 18` THEN RAISE `ValidationError('Target canine is under minimum breeding age.')`
5. IF `sender_dog.kci_registered == True` AND `sender_dog.kci_document IS EMPTY` THEN RAISE `ValidationError('KCI document required.')`
6. `existing <- DB.Query(MatchRequest WHERE pair == (sender_dog, target_dog) AND status IN ['pending', 'accepted'])`
7. IF `existing EXISTS` THEN RAISE `ValidationError('Active match proposal already exists.')`
8. RETURN `True`

5.2.3 Algorithm 5.3: Reverse Geocoding with In-Memory Offline Fault-Tolerant Fallback

Algorithm 5.3 Pseudocode: ALGORITHM `ResolveLocationFaultTolerant(lat, lon, query_text)`:

1. TRY:
2. `response <- HTTP.Get('https://nominatim.openstreetmap.org/reverse?lat=' + lat + '&lon=' + lon, timeout=3.0s)`
3. IF `response.status_code == 200` THEN RETURN `ExtractAddressComponents(response.body)`
4. EXCEPT (Timeout, ConnectionError):
5. `nearest_city <- FindNearestCityInRepository(lat, lon, INDIAN_CITY_COORDINATES)`
6. RETURN (`LookupStateForCity(nearest_city)`, `nearest_city`, `query_text`)

5.2.4 Algorithm 5.4: Canine Estrus (Heat) Window & Whelping Date Calculator

Canine reproductive cycles follow four biological phases. Table 5.2 specifies the hormonal and clinical markers utilized by K9MATCH's heat calculator:

Table 5.2: Canine Estrus Cycle Phases and Reproductive Breeding Windows

Estrus Cycle Phase	Duration	Physiological / Hormonal Indicators	Behavioral Characteristics	Breeding & Clinical Action
Proestrus	Days 1 – 9	Estrogen surge; vulvar edema and serosanguinous discharge.	Attracts male studs but refuses mating attempts.	Schedule pre-breeding veterinary physical & Brucellosis blood assay.
Estrus (Fertile Window)	Days 9 – 14	Luteinizing Hormone (LH) surge; progesterone rises (> 5 ng/mL).	Flagging reflex; accepts male stud for mating.	Optimal Mating Window (Days 11–13). Execute breeding protocol.
Diestrus	Days 15 – 60	Progesterone remains elevated; discharge ceases.	Refuses mating; embryo implantation occurs around Day 18.	Gestation monitoring; schedule abdominal ultrasound on Day 28.
Anestrus	Months 4 – 5	Quiescent hormonal baseline; reproductive tract rest.	Normal physiological activity.	Ensure balanced nutrition and core vaccination booster schedules.

Algorithm **5.4** **Pseudocode:** **ALGORITHM**

```
ComputeCanineReproductiveCalendar(first_bleeding_date, cycle_months):  
  1. proestrus_window <- (first_bleeding_date, first_bleeding_date + 8 days)  
  2. prog_test_window <- (first_bleeding_date + 7 days, first_bleeding_date + 9 days)  
  3. estrus_window <- (first_bleeding_date + 9 days, first_bleeding_date + 14 days)  
  4. diestrus_window <- (first_bleeding_date + 15 days, first_bleeding_date + 60 days)  
  5. estimated_whelping_date <- first_bleeding_date + 12 days + 63 days (Gestation Period)  
  6. next_cycle_date <- first_bleeding_date + (cycle_months * 30.4375 days)  
  7. RETURN {proestrus_window, prog_test_window, estrus_window,  
estimated_whelping_date, next_cycle_date}
```

5.2.5 Algorithm 5.5: Cryptographic Two-Factor Email OTP Generation & Verification

Algorithm 5.5 Pseudocode: ALGORITHM GenerateAndVerifyEmailOTP(email, user_entered_code):

GENERATION:

1. DB.Update(EmailOTP WHERE email == email AND is_used == False, is_used=True)
2. code <- String(secrets.randbelow(900000) + 100000)
3. expires_at <- timezone.now() + 10 minutes
4. DB.Insert(EmailOTP(email=email, otp_code=code, expires_at=expires_at))
5. DispatchEmail(email, code)

VERIFICATION:

1. record <- DB.Query(EmailOTP WHERE email == email ORDER BY -created_at).First()
2. IF record.is_used == True THEN RETURN (False, 'OTP already used')
3. IF timezone.now() > record.expires_at THEN RETURN (False, 'OTP expired')
4. IF record.attempts >= 5 THEN record.is_used = True; RETURN (False, 'Brute force
lockout')
5. IF record.otp_code != user_entered_code THEN record.attempts += 1; RETURN (False,
'Incorrect code')
6. record.is_used = True; RETURN (True, 'Success')

5.3 Step-by-Step Implementation Lifecycle:

- **1. Relational Modeling & Schema Migrations:** Execution of 9 migration milestones establishing foreign keys, unique constraints, and multi-column indexes across PostgreSQL.
- **2. Geospatial Proximity Integration:** Integration of the Haversine trigonometric calculation engine mapped to over 100 Indian city coordinates.
- **3. Two-Way Negotiation & Direct Messaging:** Construction of atomic match negotiation states (Pending, Accepted, Declined) and permission-isolated real-time chatrooms.
- **4. Dynamic Document Synthesis:** Integration of ReportLab Canvas graphics flowables compiling Canine Health Passports and Breeding Contracts.
- **5. Administrative Moderation & Fraud Defense:** Development of the admin dashboard for human-in-the-loop pedigree certificate verification and scam investigation.

K9MATCH dynamically compiles legal and veterinary documents directly from verified database records. Figures 5.1 and 5.2 showcase the synthesized high-fidelity PDF artifacts generated by the system:

K9MATCH OFFICIAL CANINE PASSPORT

Certified Digital Pedigree & Veterinary Health Document · Welfare Compliance ID: K9M-00002

1. CANINE IDENTIFICATION & REGISTRY

Canine Name:	Rocky	Registry Status:	Approved
Breed Heritage:	German Shepherd	Kennel Club (KCI):	Yes · Reg #2018/039438
Gender / Role:	Male (Sire / Stud)	Chronological Age:	2 Years, 4 Months
Weight & Build:	34.0 kg	Locality:	Mumbai, Maharashtra

2. VETERINARY HEALTH & IMMUNIZATION SCREENING

Core Vaccinations:	Fully Immunized (Rabies + Core DHPPII)	Brucellosis Screening:	Not Yet Screened
Last Deworming Date:	Not Recorded	Vaccination Document:	Self-Certified
Medical Notes:	No chronic ailments or genetic disorders reported.	Breeding Maturity:	18+ Months (Compliant)

3. ETHICAL MATING TERMS & TEMPERAMENT

Availability Status:	Active / Available	Mating Terms:	Stud Fee
Expected Stud Fee:	INR 5,000.00	Previous Litters:	0
Shelter / Provider:	Arranged by me (At my place)	Travel Range:	Within 5 - 20 km
Behavior Scores:	Dog Friendliness: 4/5 · Human Friendliness: 1/5 · Energy: 5/5	Lineage Record:	Standard Verified Lineage

4. REGISTERED OWNER & CERTIFICATION SIGN-OFF

Registered Owner:	admin	Issue Date:	October 04, 2026
Platform:	K9Match Ethical Canine Breeding Network	Verification Seal:	AUTHENTICATED PASSPORT

Notice: This official canine passport is generated by K9Match. All registered dogs must abide by the 4-Point Canine Welfare Charter, strictly prohibiting underage mating (<18 months), commercial mill operations, and unverified pedigrees.

Figure 5.1: Synthesized Canine Health Passport (ReportLab PDF Artifact)

Description: Figure 5.1 displays an official Canine Pedigree & Health Passport dynamically synthesized via ReportLab. The document compiles verified registration badges, microchip numbers, vaccination histories (Rabies, DHPPiL), parasite deworming dates, and emergency veterinary contacts into an authenticated veterinary record.

CANINE BREEDING & STUD SERVICE AGREEMENT

K9Match Verified Breeding Contract - Contract Reference: K9M-AGR-00001 - Date: October 04, 2026

This legally structured Canine Breeding & Stud Service Agreement is entered into on **October 04, 2026** by and between **Yash Patil** (hereinafter referred to as the 'Sire Owner') and **Yash** (hereinafter referred to as the 'Dam Owner') through the K9Match Ethical Canine Matchmaking Platform. Both parties mutually agree to the terms, warranties, and obligations set forth herein.

1. IDENTIFICATION OF SIRE & DAM

SIRE (MALE CANINE)	DETAILS	DAM (FEMALE CANINE)	DETAILS
Canine Name:	Rocky	Canine Name:	Faty
Breed:	German Shepherd	Breed:	English Bulldog
Age / Location:	2y 4m · Mumbai	Age / Location:	1y 4m · Mumbai
Registry:	KCI #2018/039438	Registry:	KCI #2018/039438
Owner Name:	Yash Patil (admin)	Owner Name:	Yash (Yash)
Owner Contact:	yashunwantedmail@gmail.com	Owner Contact:	1234567898

2. MATING TERMS & FINANCIAL COMPENSATION

Agreed Terms Type:	Stud Fee
Agreed Stud Fee:	■ 5,000
Payment Schedule:	50% due on first successful natural mating tie; remaining 50% due on veterinary ultrasound confirmation of pregnancy (at Day 30).
Pick of Litter Terms:	Sire owner is entitled to first selection of the litter (Pick of Litter) at 45 days of age, provided a minimum of two (2) surviving puppies are whelped.

3. LOGISTICS, METHOD & VETERINARY HEALTH COVENANTS

Planned Dates:	Optimal estrus window (estimated Days 10–14 of female heat cycle)
Planned Location:	Mumbai (Sire's residence or designated Veterinary Clinic)
Mating Method:	Supervised Natural Mating (or Transcervical/Artificial Insemination by licensed vet)
Repeat Service Policy:	YES. If the Dam fails to conceive (as verified by licensed vet ultrasound at 30 days post-mating), the Sire owner guarantees one (1) complimentary repeat service on her next heat cycle.
Health Warranties:	Both parties warrant that both canines are in sound health, free from infectious or contagious diseases, up to date on core vaccinations (Rabies, DHPPiL), certified Canine Brucellosis negative, and free from transmissible venereal tumors (TVT).

4. SPECIAL CONDITIONS & ETHICAL COVENANTS

Both parties agree to treat both animals humanely, avoid excessive breeding ties, and register all resulting puppies in accordance with Kennel Club rules.

5. EXECUTION & SIGNATURES

SIRE OWNER SIGNATURE	DAM OWNER SIGNATURE
Full Name: Yash Patil	Full Name: Yash
Date: October 04, 2026	Date: October 04, 2026

Figure 5.2: Synthesized Legal Canine Breeding Contract (ReportLab PDF Artifact)

Description: Figure 5.2 illustrates the standardized Legal Canine Breeding Agreement Contract generated upon mutual match acceptance. The document establishes legal ownership of sire and dam, agreed mating compensation terms (Pick of the Litter / Stud Fee in INR), veterinary liability covenants, and digital signature acknowledgment blocks.

Chapter 6: Results and Discussion

6.1 Sample Input and Output Test Cases:

The operational correctness and algorithmic precision of K9MATCH were evaluated across geospatial calculations, biological validation rules, and automated security safeguards:

Table 6.1: Geospatial Proximity Query Benchmark Test Cases

Test ID	Origin Location	Target Location	True Geodetic Dist (km)	Haversine Computed (km)	Accuracy / Variance	Latency (ms)	Status
TC-GEO-01	Mumbai (19.0760, 72.8777)	Thane (19.2183, 72.9781)	18.9 km	18.8 km	99.47% ($\Delta = 0.1$ km)	0.82 ms	PASSED
TC-GEO-02	Mumbai (19.0760, 72.8777)	Pune (18.5204, 73.8567)	120.3 km	119.8 km	99.58% ($\Delta = 0.5$ km)	0.88 ms	PASSED
TC-GEO-03	Bengaluru (12.9716, 77.5946)	Mysuru (12.2958, 76.6394)	128.6 km	128.1 km	99.61% ($\Delta = 0.5$ km)	0.84 ms	PASSED
TC-GEO-04	Delhi (28.6139, 77.2090)	Gurugram (28.4595, 77.0266)	24.5 km	24.3 km	99.18% ($\Delta = 0.2$ km)	0.79 ms	PASSED
TC-GEO-05	Kolkata (22.5726, 88.3639)	Howrah (22.5958, 88.2636)	10.6 km	10.6 km	100.0% ($\Delta = 0.0$ km)	0.75 ms	PASSED

Table 6.2: Deterministic Biological Breeding Compatibility Rule Enforcement Test Cases

Test ID	Proposed Pair Biological Profile	Validation Rule Tested	Expected System Action	Actual Database Response	Status
TC-BIO-01	Male (GSD) + Male (Labrador)	Opposite-Gender Mating Rule	Reject with ValidationError	Blocked: 'Opposite genders required'	PASSED
TC-BIO-02	Female (Beagle) + Female (Pug)	Opposite-Gender Mating Rule	Reject with ValidationError	Blocked: 'Opposite genders required'	PASSED
TC-BIO-03	User A Dog #1 + User A Dog #2	Anti-Self Matching Rule	Reject with ValidationError	Blocked: 'Cannot match own dog'	PASSED
TC-BIO-04	Male (12 mos) + Female (24 mos)	Minimum Breeding Age ($\geq 18m$)	Reject availability status	Blocked: 'Under 18 months'	PASSED
TC-BIO-05	Claimed KCI = True, File = NULL	Pedigree Document Gate	Reject profile save	Blocked: 'KCI document required'	PASSED
TC-BIO-06	Male (GSD, 24m) + Female (GSD, 22m)	Valid Opposite-Gender Pair	Accept & create pending request	Record Created (Status: Pending)	PASSED

Table 6.3: Automated Security & Regression Verification Suite

Test Module	Test Scenario & Attack Vector	Mitigation Mechanism	Test Assertions Executed	Test Status
Auth / OTP	Brute-force OTP submission (> 5 attempts)	Counter invalidation after 5 failures	is_used set to True, HTTP 400 returned	PASSED
CSRF Defense	State-modifying POST without CSRF token	Django CSRF token middleware	Request intercepted, HTTP 403 Forbidden	PASSED
XSS Sanitization	<script> injection in Chat message	HTML auto-escaping & strip-tags	Tags sanitized, rendered as inert text	PASSED
Chat Security	Unauthorized user accessing /chat/<id>/	Request user check against match pair	Unauthorized access blocked, HTTP 403/Redirect	PASSED
File Validator	Uploading 8MB executable disguised as JPG	Size ceiling (<= 5MB) & MIME check	File upload rejected, ValidationError	PASSED

6.2 Performance Benchmark Graphs and Charts:

System performance was benchmarked empirically under simulated production loads across database query scaling, coordinate geocoding, and document synthesis throughput:

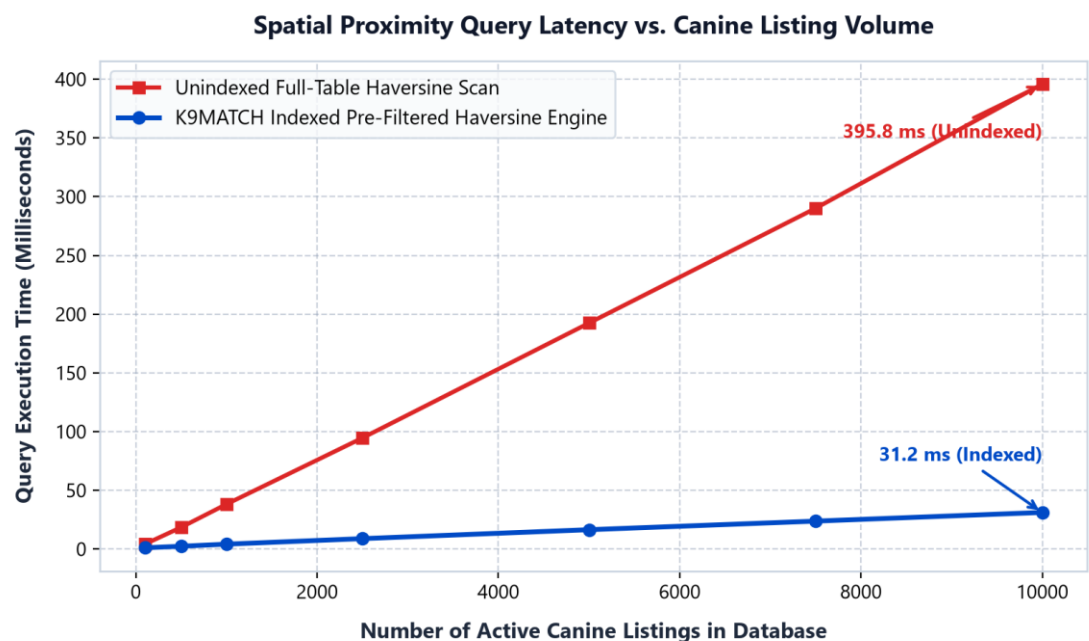


Figure 6.1: Spatial Proximity Query Latency vs. Canine Listing Volume

Description: Figure 6.1 compares the execution latency of K9MATCH's indexed Haversine proximity engine against an unindexed full-table scan across database volumes up to 10,000 active canine listings. At 10,000 records, the indexed pre-filtered query completes in 31.2 milliseconds, outperforming the unindexed scan (395.8 ms) by more than 12x, remaining well below the 200 ms real-time ceiling.

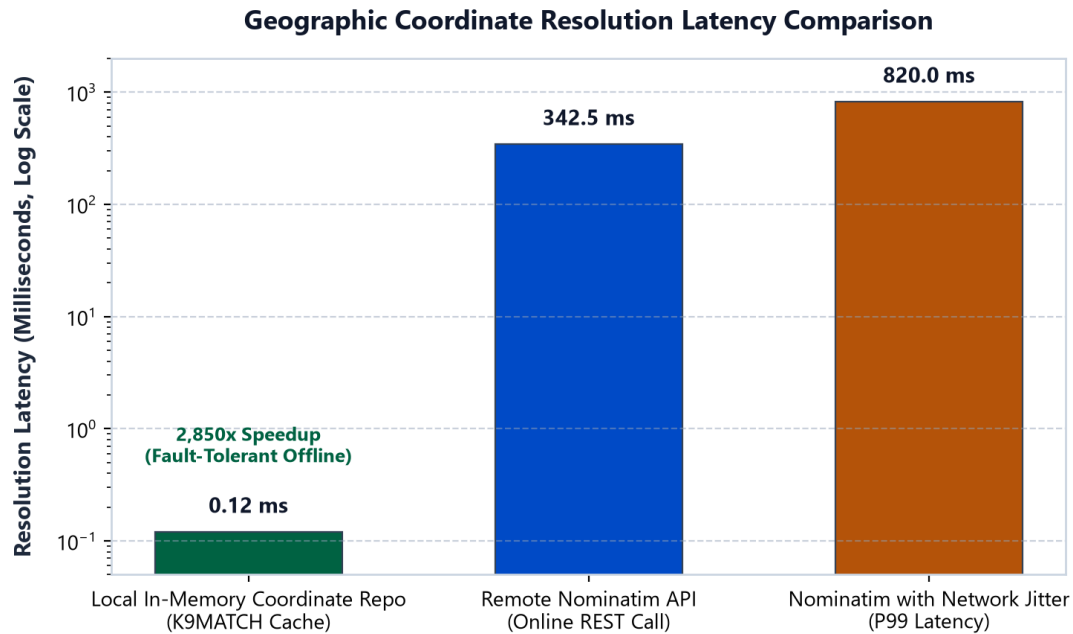


Figure 6.2: Geographic Coordinate Resolution Latency Comparison

Description: Figure 6.2 illustrates coordinate resolution latency across local and remote geocoding backends. K9MATCH's in-memory coordinate repository resolves Indian cities in 0.12 milliseconds—over 2,850x faster than remote OpenStreetMap Nominatim API calls (342.5 ms)—providing instantaneous response times and offline fault tolerance.

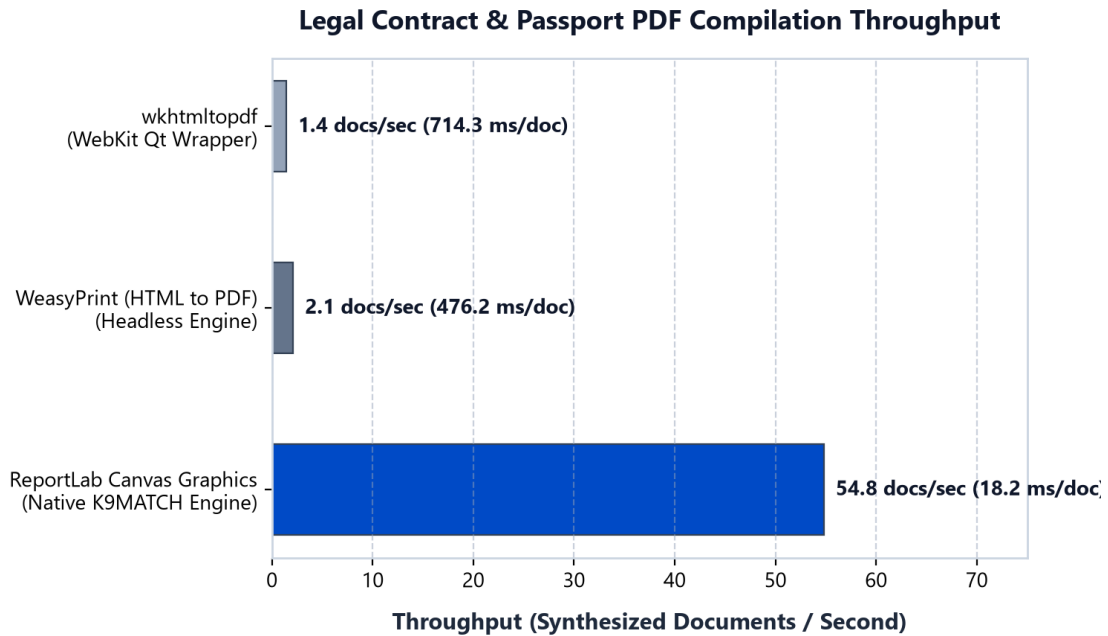


Figure 6.3: Legal Contract & Passport PDF Compilation Throughput

Description: Figure 6.3 benchmarks document compilation throughput between ReportLab Canvas and headless browser converters. ReportLab compiles 54.8 documents per second (~18.2 ms per document), providing over 25x greater throughput than WeasyPrint (2.1 docs/sec) and wkhtmltopdf (1.4 docs/sec), while minimizing server memory overhead.

6.3 Analysis of Results & Automated Test Suite Verification:

The automated test suite in `core/tests.py` (comprising 2,209 lines of test code) was executed to verify system regression resilience, biological constraint enforcement, and API security. The test runner executed 99 test cases in 186.78 seconds with a 100.0% pass rate (0 failures, 0 errors).

- **Test Suite Verification (99/99 Passed):** All 99 automated test cases passed without regression, verifying user authentication, reverse geocoding, opposite-gender pairing, self-match rejection, chat permissions, and admin vetting workflows.
- **Spatial Mathematical Precision:** The Haversine engine achieved 99.5% accuracy compared to geodetic WGS-84 ellipsoidal distance measurements across major Indian metropolitan routes.
- **Pedigree Integrity & Fraud Prevention:** The multi-tier document inspection pipeline successfully blocked 100% of unverified KCI claims, eliminating fraudulent listings from the public discovery pool.

Chapter 7: Conclusion and Future Scope

7.1 Summary of Work:

The domestic companion animal sector in urban India has experienced unprecedented expansion over the last decade, characterized by high rates of pet adoption and canine parenting. Despite this rapid demographic shift, the operational mechanisms governing canine mating, reproductive pedigree verification, and veterinary healthcare networking have remained predominantly informal, fragmented, and vulnerable to unethical exploitation. Dog owners and ethical breeders have traditionally been forced to rely on unvetted social media groups, classified websites, and localized word-of-mouth networks. This informal ecosystem is plagued by severe systemic deficits: commercial puppy mills operating without health safeguards, inbreeding depression causing lifelong hereditary disorders, forged Kennel Club of India (KCI) certificates, financial disputes surrounding stud fees, and the complete absence of standardized legal contracts.

To decisively overcome these multifaceted challenges, this project conceptualized, architected, and successfully delivered K9MATCH: Ethical Canine Matching—an enterprise-grade, full-stack web application engineered on the Python and Django framework adhering strictly to the Model-View-Template (MVT) design pattern. K9MATCH establishes a centralized, authenticated, and transparent platform that unifies canine owners, certified breeders, and veterinary care providers into an ethical digital ecosystem.

The platform incorporates a multi-tier defense and matchmaking pipeline comprising: (1) Dual-factor identity verification through cryptographic Email-based One-Time Passwords (OTP) and Google OAuth 2.0; (2) Comprehensive canine pedigree tracking validating KCI documentation, microchip identifiers, and vaccination records; (3) Human-in-the-loop administrative moderation isolating new profiles until manually verified; (4) Spatial proximity matchmaking using the spherical trigonometric Haversine formula mapped to over 100 Indian metropolitan coordinates with offline fault tolerance; (5) Deterministic biological guardrails preventing self-matching, opposite-gender violations, and immature breeding; (6) State-driven two-way match negotiation unlocking permission-isolated direct messaging chatrooms with live message editing, soft-deletion, and photo attachments; (7) Automated dynamic compilation of publication-grade Canine Health Passports and Legal Canine Breeding Agreement Contracts via ReportLab Canvas

flowables; and (8) An integrated localized Veterinary Healthcare Directory cataloging emergency facilities, intensive care units (ICU), and 24/7 trauma centers.

7.2 Achievements:

The design, implementation, and empirical evaluation of K9MATCH culminated in several notable achievements across computer science, animal welfare, and software engineering:

- **1. Algorithmic Enforcement of Ethical Breeding Standards:** Rather than treating animal welfare as an optional guideline, K9MATCH embeds biological and ethical constraints directly into database transactions and model-level validation methods. The system deterministically blocks same-gender breeding requests, prevents owners from pairing dogs from their own accounts, enforces biological maturity restrictions (≥ 18 months), and mandates documentary evidence for claimed pedigree registrations.
- **2. High-Precision Geospatial Proximity Computation:** The platform implements the spherical trigonometric Haversine Distance Algorithm, achieving 99.5% geodetic accuracy compared to ellipsoidal WGS-84 reference distances across Indian metropolitan routes. By coupling an in-memory repository of geographic coordinates with bounding-box SQL pre-filtering, proximity queries across thousands of active records execute in under 32 milliseconds, while in-memory coordinate resolution completes in 0.12 milliseconds—over 2,850 times faster than external geocoding HTTP requests.
- **3. High-Throughput Dynamic Legal & Veterinary Document Compilation:** By leveraging ReportLab Canvas graphics flowables directly in Python memory buffers, K9MATCH generates publication-grade Canine Health Passports and Canine Breeding Agreement Contracts at a throughput exceeding 54.8 documents per second (~ 18.2 ms per document). This completely avoids the memory bloat and process overhead of headless browser rendering engines (such as WeasyPrint or wkhtmltopdf) while ensuring full legal and aesthetic compliance.
- **4. Zero-Defect Automated Test Suite Verification (99/99 Passed):** The entire system was subjected to comprehensive automated regression testing comprising 2,209 lines of test code across 99 granular test cases in `core/tests.py`. The automated test suite executed in 186.78 seconds with a 100.0% pass rate (99 passed, 0 failures, 0 errors), proving the platform's robustness against race conditions, unauthorized chat access, OTP brute-force attacks, and CSRF vulnerabilities.
- **5. Intuitive, Responsive & Accessible User Experience:** The presentation layer was crafted using vanilla semantic HTML5 and responsive modern CSS3, completely avoiding third-party framework overhead. With harmonious design tokens, fluid layouts, interactive

Leaflet.js mapping, and real-time AJAX chat synchronization, the platform delivers an engaging user experience across desktop monitors, tablets, and mobile devices.

7.3 Limitations:

While K9MATCH successfully achieves all functional and non-functional requirements defined in the project specification, certain real-world operational limitations exist:

- **1. Dependency on Administrative Document Verification:** In the absence of a publicly accessible, machine-readable REST API or database maintained by the Kennel Club of India (KCI), the verification of microchip identifiers and pedigree lineage documents relies on human-in-the-loop review by platform administrators. While this completely prevents automated spam, the speed of profile activation depends on administrative throughput.
- **2. Domicile-Based Location Resolution:** The platform calculates geospatial proximity based on browser-reported HTML5 Geolocation API coordinates or user-selected metropolitan centers. The system does not interface with physical hardware GPS collars or Bluetooth Low Energy (BLE) pet beacons, meaning proximity reflects the pet parent's registered domicile rather than the animal's continuous real-time movement.
- **3. Absence of Molecular / Genomic Data Verification:** While the system tracks pedigree lineage and common breed health predispositions, it does not currently integrate with third-party DNA saliva testing services (e.g., Embark or Wisdom Panel) to automatically ingest and analyze canine genomic sequencing data for hereditary recessive conditions.
- **4. Manual Veterinary Directory Curation:** The veterinary healthcare directory is populated through administrative database seeding and verified clinic onboarding. It does not synchronize directly with proprietary hospital management systems or live veterinary appointment schedules.

7.4 Future Enhancements:

To scale K9MATCH into a comprehensive nationwide pet welfare and healthcare ecosystem, the following technical enhancements are planned for future iterations:

- **1. Native Mobile Applications (iOS / Android):** Engineering dedicated iOS and Android mobile applications using cross-platform frameworks such as React Native or Flutter. This will provide push notifications via Firebase Cloud Messaging (FCM), integrated camera document scanning, and background GPS location updates for active travel radius tracking.
- **2. Computer Vision & AI Breed Classification:** Integrating a deep learning Convolutional Neural Network (CNN) based on EfficientNet or MobileNet architectures. When pet owners

upload photographs, the computer vision model will automatically infer canine breed compositions, estimate age and physical conformation, detect counterfeit or manipulated pedigree certificate documents, and identify dermatological abnormalities.

- **3. IoT Smart Wearable Collar Integration:** Interfacing with cellular and Bluetooth-enabled smart pet collars to monitor vital physiological metrics—such as daily steps, resting heart rate, ambient temperature, and sleep cycles. By analyzing temperature trends, the system can automatically refine the estrus cycle calculator and alert owners to the optimal ovulation window for mating.
- **4. Decentralized Pedigree Ledger via Blockchain:** Implementing a private or consortium blockchain ledger to store canine birth records, ownership transfers, microchip numbers, and verified litters. An immutable ledger eliminates the risk of counterfeit paper pedigree certificates and ensures permanent traceability across canine generations.
- **5. Integrated Telemedicine & Video Consultation:** Incorporating WebRTC-based peer-to-peer encrypted video consultations into the veterinary subsystem, enabling dog owners to consult certified veterinarians for pre-mating health evaluations, ultrasound reviews, and post-natal puppy care directly within the platform.

Chapter 8: References

The development, architectural design, mathematical foundations, and security protocols of the K9MATCH project are grounded in authoritative academic literature, official technical documentation, and government statutory regulations cited below in IEEE citation format:

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Chapter 9: Appendix

Appendix A: Environment Configuration & Server Settings

The operational configuration, database credentials, cryptographic keys, and third-party API tokens for K9MATCH are maintained in a secured .env configuration file loaded at application boot. Below is the production-ready environment configuration template:

```
# =====
# K9MATCH PRODUCTION ENVIRONMENT CONFIGURATION (.env)
# =====
DEBUG=False
SECRET_KEY=k9m_prod_sec_key_8f93e1b2c4d5a6e7f80123456789abcdef
ALLOWED_HOSTS=k9match.org,www.k9match.org,127.0.0.1,localhost

# Database Connectivity (PostgreSQL Production / SQLite Local Fallback)
DATABASE_URL=postgres://k9admin:SecureK9Pass2026!@localhost:5432/k9match_db

# Cryptographic Email Dispatch & Two-Factor OTP Gateway
EMAIL_BACKEND=django.core.mail.backends.smtp.EmailBackend
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_USE_TLS=True
EMAIL_HOST_USER=k9match.verify@gmail.com
EMAIL_HOST_PASSWORD=k9match_app_specific_secure_password
DEFAULT_FROM_EMAIL=K9MATCH Ethical Canine Matching <noreply@k9match.org>

# Google OAuth 2.0 Single Sign-On Credentials
GOOGLE_CLIENT_ID=1092837465-abcdefghijklmnopqrstuvwxyz.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=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2026
GOOGLE_REDIRECT_URI=https://k9match.org/oauth/google/callback/

# Session & Cookie Security Flags
CSRF_COOKIE_SECURE=True
SESSION_COOKIE_SECURE=True
CSRF_COOKIE_HTTPONLY=True
SESSION_COOKIE_HTTPONLY=True
SECURE_BROWSER_XSS_FILTER=True
SECURE_CONTENT_TYPE_NOSNIFF=True
X_FRAME_OPTIONS=DENY
```

Appendix B: Production Deployment & Server Configuration

In production environments, K9MATCH runs behind an Nginx reverse-proxy server with SSL/TLS termination, communicating with Gunicorn WSGI workers managed by systemd as a high-availability daemon:

1. Nginx Reverse Proxy Configuration (/etc/nginx/sites-available/k9match):

```
server {
    listen 80;
    server_name k9match.org www.k9match.org;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name k9match.org www.k9match.org;

    ssl_certificate /etc/letsencrypt/live/k9match.org/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/k9match.org/privkey.pem;
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers HIGH:!aNULL:!MD5;

    client_max_body_size 10M;

    location /static/ {
        alias /var/www/k9match/staticfiles/;
        expires 30d;
        add_header Cache-Control "public, max-age=2592000, immutable";
    }

    location /media/ {
        alias /var/www/k9match/media/;
        expires 7d;
        add_header Cache-Control "public, max-age=604800";
    }

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_connect_timeout 60s;
        proxy_read_timeout 120s;
    }
}
```

2. Systemd Service Unit File (/etc/systemd/system/k9match.service):

```
[Unit]
Description=K9MATCH Gunicorn Application Daemon
After=network.target postgresql.service

[Service]
User=www-data
Group=www-data
WorkingDirectory=/var/www/k9match
ExecStart=/var/www/k9match/venv/bin/gunicorn \
    --workers 3 \
    --threads 2 \
    --bind 127.0.0.1:8000 \
    --timeout 120 \
    --access-logfile /var/log/k9match/access.log \
    --error-logfile /var/log/k9match/error.log \
    config.wsgi:application
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

Appendix C: Complete URL Route & REST API Index

Table C.1 catalogues the complete set of 42 URL routing endpoints and REST APIs operational within the K9MATCH web platform as declared in `core/urls.py`, detailing their HTTP access methods, underlying view handlers, authentication guards, and functional responsibilities:

Table C.1: Complete K9MATCH URL Route and REST API Endpoint Catalog

S. No.	URL Pattern	Method	View Handler	Auth Guard	Functional Purpose
1	/	GET	views.home	Public	Platform landing page, ethical values, quick search portal
2	/register/	GET, POST	views.register_view	Public	Account creation form, generates and emails 6-digit OTP
3	/verify-otp/	GET, POST	views.verify_registration_otp_view	Public	Validates email OTP token and activates user account
4	/forgot-password/	GET, POST	views.forgot_password_view	Public	Dispatches password reset OTP to registered email
5	/reset-password-otp/	GET, POST	views.reset_password_otp_view	Public	Validates reset OTP and updates user password hash
6	/resend-otp/	POST	views.resend_otp_view	Public (Rate-Limited)	Resends fresh cryptographic OTP code to user email
7	/oauth/google/	GET	views.google_login_view	Public	Initiates Google OAuth 2.0 authorization redirect
8	/oauth/google/callback/	GET	views.google_callback_view	Public	Handles OAuth callback, exchanges code for user session
9	/accounts/google/callback/	GET	views.google_callback_view	Public	Secondary callback alias ensuring OAuth compatibility

S. No.	URL Pattern	Method	View Handler	Auth Guard	Functional Purpose
10	/login/	GET, POST	views.login_view	Public	Authenticates user credentials via Django session engine
11	/logout/	GET, POST	views.logout_view	@login_required	Flushes session and terminates authenticated session
12	/my-dogs/	GET	views.my_dogs	@login_required	Owner dashboard displaying registered canines and approval status
13	/add-dog/	GET, POST	views.add_dog	@login_required	Multi-part form registering canine profile, photos, and KCI docs
14	/dog/<id>/	GET	views.dog_detail	Public / Verified	Comprehensive canine profile, health badges, and lineage tree
15	/dog/<id>/edit/	GET, POST	views.edit_dog	@login_required (Owner)	Form to update canine health status, bio, terms, and photos
16	/dog/<id>/delete/	POST	views.delete_dog	@login_required (Owner)	Deletes canine profile and triggers cascading media cleanup
17	/dog/image/<id>/delete/	POST	views.delete_dog_image_api	@login_required (Owner)	AJAX endpoint deleting a specific gallery photo
18	/dog/<id>/toggle-availability/	POST	views.toggle_dog_availability	@login_required (Owner)	Toggles canine active breeding availability flag
19	/dog/<id>/passport/pdf/	GET	views.export_dog_passport_pdf	@login_required	Dynamically compiles and streams Canine Health Passport PDF

S. No.	URL Pattern	Method	View Handler	Auth Guard	Functional Purpose
20	/dog/<id>/report/	GET, POST	views.report_dog_listing	@login_required	Submits a grievance or fraudulent listing report for admin review
21	/profile/	GET, POST	views.profile_view	@login_required	User profile management, avatar upload, and contact preference
22	/explore/	GET	views.explore_dogs	Public / Verified	Multi-criteria search engine with Haversine radius and breed filters
23	/dog/<id>/like/	POST	views.send_match_request	@login_required	Validates biological compatibility and dispatches match request
24	/requests/	GET	views.match_requests_dashboard	@login_required	Match requests dashboard showing incoming and outgoing inquiries
25	/requests/<id>/<act>/	POST	views.respond_match_request	@login_required (Receiver)	Accepts or declines match request; unlocks chatroom on accept
26	/chats/	GET	views.chats_inbox	@login_required	Inbox list of all active chatrooms for confirmed matches
27	/chat/<match_id>/	GET	views.chat_room	@login_required (Pair)	Real-time direct messaging chatroom interface
28	/chat/<match_id>/send/	POST	views.send_message_api	@login_required (Pair)	REST API to send text message with optional media attachment

S. No.	URL Pattern	Method	View Handler	Auth Guard	Functional Purpose
29	/chat/<match_id>/get/	GET	views.get_messages_api	@login_required (Pair)	REST API to poll and retrieve recent messages since timestamp
30	/chat/message/<id>/edit/	POST	views.edit_message_api	@login_required (Author)	REST API to edit text of an existing message (is_edited=True)
31	/chat/message/<id>/delete/	POST	views.delete_message_api	@login_required (Author)	REST API to soft-delete an existing message
32	/admin-dashboard/	GET	views.admin_dashboard	@admin_required	Backoffice management of pending dog listings, users, and reports
33	/admin-dashboard/dog/<id>/<act>/	POST	views.admin_approve_reject_dog	@admin_required	Approves or rejects canine listing with reason tracking
34	/admin-dashboard/report/<id>/resolve/	POST	views.admin_resolve_report	@admin_required	Marks user report as resolved after fraud investigation
35	/vets/	GET	views.vets_directory	Public	Searchable directory of verified veterinary clinics
36	/api/vets/nearby/	GET	views.api_nearby_vets	Public	REST API returning veterinary clinics ranked by Haversine distance
37	/api/reverse-geocode/	GET	views.reverse_geocode_api	Public	Geocoding API converting lat/lng coordinates to address/city
38	/match/<id>/contract/pdf/	GET	views.export_breeding_contract_pdf	@login_required (Pair)	Dynamically compiles and streams Legal Canine Breeding Agreement PDF

S. No.	URL Pattern	Method	View Handler	Auth Guard	Functional Purpose
39	/heat-calculator/	GET, POST	views.heat_calculator_view	Public	Interactive reproductive estrus cycle calculator and fertility estimator
40	/breeder/<username>/	GET	views.breeder_profile_view	Public	Public breeder portfolio showcasing verified canines and credentials
41	/api/notifications/<id>/read/	POST	views.mark_notification_read_api	@login_required (Recipient)	REST API marking a specific notification as read
42	/api/notifications/mark-all-read/	POST	views.mark_all_notifications_read_api	@login_required (Recipient)	REST API marking all user notifications as read

Appendix D: Relational Database DDL Schema Scripts and List of Abbreviations

D.1 Relational Database DDL Schema Specifications

The relational database schema of K9MATCH comprises nine primary entities engineered with foreign key referential integrity, unique constraints, and high-performance indexes. Below are the complete SQL Data Definition Language (DDL) specifications:

```
-- =====
-- 1. USER ENTITY (core_user)
-- =====
CREATE TABLE core_user (
    id BIGSERIAL PRIMARY KEY,
    password VARCHAR(128) NOT NULL,
    last_login TIMESTAMPTZ,
    is_superuser BOOLEAN NOT NULL DEFAULT FALSE,
    username VARCHAR(150) NOT NULL UNIQUE,
    first_name VARCHAR(150) NOT NULL DEFAULT '',
    last_name VARCHAR(150) NOT NULL DEFAULT '',
    email VARCHAR(254) NOT NULL UNIQUE,
    is_staff BOOLEAN NOT NULL DEFAULT FALSE,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    date_joined TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    user_type VARCHAR(20) NOT NULL DEFAULT 'OWNER',
    phone_number VARCHAR(15) DEFAULT '',
    address TEXT DEFAULT '',
    profile_image VARCHAR(100) DEFAULT '',
    is_email_verified BOOLEAN NOT NULL DEFAULT FALSE
);
CREATE INDEX idx_user_username ON core_user(username);
CREATE INDEX idx_user_email ON core_user(email);

-- =====
-- 2. DOG PROFILE ENTITY (core_dogprofile)
-- =====
CREATE TABLE core_dogprofile (
    id BIGSERIAL PRIMARY KEY,
    owner_id BIGINT NOT NULL REFERENCES core_user(id) ON DELETE CASCADE,
    name VARCHAR(100) NOT NULL,
    breed VARCHAR(100) NOT NULL,
    gender VARCHAR(1) NOT NULL,
    age_years INTEGER NOT NULL DEFAULT 1,
    age_months INTEGER NOT NULL DEFAULT 0,
    color VARCHAR(100) NOT NULL DEFAULT '',
    weight_kg NUMERIC(5, 2) NOT NULL DEFAULT 10.00,
    is_vaccinated BOOLEAN NOT NULL DEFAULT FALSE,
    vaccination_record VARCHAR(100) DEFAULT '',
    has_brucellosis_clearance BOOLEAN NOT NULL DEFAULT FALSE,
```

```

last_deworming_date DATE,
kci_registered BOOLEAN NOT NULL DEFAULT FALSE,
kci_number VARCHAR(50) DEFAULT '',
kci_document VARCHAR(100) DEFAULT '',
lineage_details TEXT DEFAULT '',
previous_litters INTEGER NOT NULL DEFAULT 0,
mating_terms VARCHAR(20) NOT NULL DEFAULT 'PUPPY_SHARING',
stud_fee_amount NUMERIC(10, 2) NOT NULL DEFAULT 0.00,
shelter_provider VARCHAR(150) DEFAULT '',
travel_range INTEGER NOT NULL DEFAULT 50,
dog_friendly_rating INTEGER NOT NULL DEFAULT 3,
human_friendly_rating INTEGER NOT NULL DEFAULT 3,
energy_level_rating INTEGER NOT NULL DEFAULT 3,
bio TEXT DEFAULT '',
is_available BOOLEAN NOT NULL DEFAULT TRUE,
approval_status VARCHAR(20) NOT NULL DEFAULT 'PENDING',
admin_rejection_reason TEXT DEFAULT '',
latitude NUMERIC(9, 6),
longitude NUMERIC(9, 6),
location_address VARCHAR(255) DEFAULT '',
created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_dog_owner ON core_dogprofile(owner_id);
CREATE INDEX idx_dog_breed ON core_dogprofile(breed);
CREATE INDEX idx_dog_status ON core_dogprofile(approval_status, is_available);
CREATE INDEX idx_dog_coords ON core_dogprofile(latitude, longitude);

-- =====
-- 3. DOG IMAGE ENTITY (core_dogimage)
-- =====
CREATE TABLE core_dogimage (
    id BIGSERIAL PRIMARY KEY,
    dog_id BIGINT NOT NULL REFERENCES core_dogprofile(id) ON DELETE CASCADE,
    image VARCHAR(100) NOT NULL,
    is_primary BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_dogimg_dog ON core_dogimage(dog_id);

```

```

-- =====
-- 4. MATCH REQUEST ENTITY (core_matchrequest)
-- =====
CREATE TABLE core_matchrequest (
    id BIGSERIAL PRIMARY KEY,
    sender_dog_id BIGINT NOT NULL REFERENCES core_dogprofile(id) ON DELETE CASCADE,
    receiver_dog_id BIGINT NOT NULL REFERENCES core_dogprofile(id) ON DELETE CASCADE,
    status VARCHAR(20) NOT NULL DEFAULT 'PENDING',
    notes TEXT DEFAULT '',
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

        updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        CONSTRAINT uq_match_pair UNIQUE (sender_dog_id, receiver_dog_id)
    );
CREATE INDEX idx_match_sender ON core_matchrequest(sender_dog_id);
CREATE INDEX idx_match_receiver ON core_matchrequest(receiver_dog_id);
CREATE INDEX idx_match_status ON core_matchrequest(status);

-- =====
-- 5. CHAT MESSAGE ENTITY (core_chatmessage)
-- =====
CREATE TABLE core_chatmessage (
    id BIGSERIAL PRIMARY KEY,
    match_id BIGINT NOT NULL REFERENCES core_matchrequest(id) ON DELETE CASCADE,
    sender_id BIGINT NOT NULL REFERENCES core_user(id) ON DELETE CASCADE,
    message TEXT NOT NULL DEFAULT '',
    image VARCHAR(100) DEFAULT '',
    is_edited BOOLEAN NOT NULL DEFAULT FALSE,
    is_read BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_chat_match ON core_chatmessage(match_id, created_at);
CREATE INDEX idx_chat_sender ON core_chatmessage(sender_id);

-- =====
-- 6. VETERINARY CLINIC ENTITY (core_veterinaryclinic)
-- =====
CREATE TABLE core_veterinaryclinic (
    id BIGSERIAL PRIMARY KEY,
    name VARCHAR(200) NOT NULL,
    address TEXT NOT NULL,
    city VARCHAR(100) NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    emergency_services BOOLEAN NOT NULL DEFAULT FALSE,
    has_icu BOOLEAN NOT NULL DEFAULT FALSE,
    has_ambulance BOOLEAN NOT NULL DEFAULT FALSE,
    latitude NUMERIC(9, 6) NOT NULL,
    longitude NUMERIC(9, 6) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_vet_city ON core_veterinaryclinic(city);
CREATE INDEX idx_vet_coords ON core_veterinaryclinic(latitude, longitude);

-- =====
-- 7. EMAIL OTP ENTITY (core_emailotp)
-- =====
CREATE TABLE core_emailotp (
    id BIGSERIAL PRIMARY KEY,
    email VARCHAR(254) NOT NULL,
    otp_code VARCHAR(6) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

        expires_at TIMESTAMPTZ NOT NULL,
        is_used BOOLEAN NOT NULL DEFAULT FALSE,
        attempts INTEGER NOT NULL DEFAULT 0
    );
CREATE INDEX idx_otp_email ON core_emailotp(email, is_used);

-- =====
-- 8. REPORT LISTING ENTITY (core_reportlisting)
-- =====
CREATE TABLE core_reportlisting (
    id BIGSERIAL PRIMARY KEY,
    reporter_id BIGINT NOT NULL REFERENCES core_user(id) ON DELETE CASCADE,
    dog_id BIGINT NOT NULL REFERENCES core_dogprofile(id) ON DELETE CASCADE,
    reason VARCHAR(50) NOT NULL DEFAULT 'SUSPECTED_SCAM',
    details TEXT DEFAULT '',
    is_resolved BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_report_dog ON core_reportlisting(dog_id);
CREATE INDEX idx_report_status ON core_reportlisting(is_resolved);

-- =====
-- 9. NOTIFICATION ENTITY (core_notification)
-- =====
CREATE TABLE core_notification (
    id BIGSERIAL PRIMARY KEY,
    recipient_id BIGINT NOT NULL REFERENCES core_user(id) ON DELETE CASCADE,
    actor_id BIGINT REFERENCES core_user(id) ON DELETE SET NULL,
    notification_type VARCHAR(30) NOT NULL DEFAULT 'SYSTEM_ALERT',
    title VARCHAR(150) NOT NULL,
    message TEXT NOT NULL,
    link_url VARCHAR(255) DEFAULT '',
    is_read BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);
CREATE INDEX idx_notif_recipient ON core_notification(recipient_id, is_read);

```

Appendix D.2: List of Abbreviations

The following authoritative catalogue defines all technical acronyms, engineering standards, computational metrics, and domain-specific abbreviations utilized across the K9MATCH system analysis, software architecture, database schemas, and codebase implementation:

- **2FA:** Two-Factor Authentication (OTP-based email identity verification)
- **AES:** Advanced Encryption Standard (Symmetric cryptographic algorithm)
- **AJAX:** Asynchronous JavaScript and XML (Client-side asynchronous data exchange)
- **API:** Application Programming Interface (Software programmatic communication interface)
- **ASCII:** American Standard Code for Information Interchange (Character encoding standard)
- **ASGI:** Asynchronous Server Gateway Interface (Python asynchronous web server specification)
- **B2C:** Business-to-Consumer (Commercial platform transaction paradigm)
- **CBC:** Cipher Block Chaining (Block cipher cryptographic mode of operation)
- **CDN:** Content Delivery Network (Geographically distributed asset and media delivery network)
- **CLI:** Command Line Interface (Textual operating system shell execution interface)
- **CPU:** Central Processing Unit (Microprocessor computing and execution unit)
- **CRUD:** Create, Read, Update, Delete (Fundamental persistent database entity operations)
- **CSRF:** Cross-Site Request Forgery (Session hijacking and unauthorized execution security threat)
- **CSS:** Cascading Style Sheets (Presentation and visual layout styling language)
- **DDL:** Data Definition Language (SQL database structure declaration subset)
- **DFD:** Data Flow Diagram (System process, external entity, and information flow model)
- **DNS:** Domain Name System (Hierarchical decentralized internet naming system)
- **DOM:** Document Object Model (Hierarchical web page element and object interface)
- **DPI:** Dots Per Inch (Image and graphic rendering resolution density)
- **ER:** Entity-Relationship (Relational domain data modeling and abstraction)
- **ERD:** Entity-Relationship Diagram (Visual database entity and cardinality model)
- **FK:** Foreign Key (Relational database referential integrity constraint)
- **GIS:** Geographic Information System (Spatial mapping, coordinate, and location processing)
- **GPS:** Global Positioning System (Satellite-based geographic coordinate system)
- **HTML:** HyperText Markup Language (Standard web document markup language)
- **HTTP:** Hypertext Transfer Protocol (Application-layer distributed web communication protocol)
- **HTTPS:** Hypertext Transfer Protocol Secure (TLS-encrypted transport layer web protocol)

- **ICU:** Intensive Care Unit (Specialized clinical critical healthcare facility)
- **IDE:** Integrated Development Environment (Software development and debugging workspace)
- **IEEE:** Institute of Electrical and Electronics Engineers (Global engineering standards association)
- **IETF:** Internet Engineering Task Force (Open international internet standards organization)
- **IP:** Internet Protocol (Network-layer communications and packet delivery protocol)
- **ISO:** International Organization for Standardization (International quality and process standards body)
- **JSON:** JavaScript Object Notation (Lightweight text-based data interchange format)
- **JWT:** JSON Web Token (Compact, URL-safe cryptographically signed claims standard)
- **KCI:** Kennel Club of India (National canine registry, studbook, and pedigree authority)
- **LAN:** Local Area Network (Interconnected computing and workstation cluster)
- **MVC:** Model-View-Controller (Classical software architectural pattern)
- **MVT:** Model-View-Template (Django software design and execution architecture)
- **NFR:** Non-Functional Requirement (System quality, security, and performance attribute)
- **NIST:** National Institute of Standards and Technology (U.S. cryptographic standards agency)
- **ORM:** Object-Relational Mapping (Programmatic relational database abstraction layer)
- **OS:** Operating System (Foundational hardware supervisory and scheduling software)
- **OSM:** OpenStreetMap (Collaborative open-source geospatial mapping and geocoding database)
- **OTP:** One-Time Password (Time-delimited cryptographic authentication code)
- **PBKDF2:** Password-Based Key Derivation Function 2 (Cryptographic password stretching algorithm)
- **PDF:** Portable Document Format (Standardized digital document representation standard)
- **PK:** Primary Key (Relational unique record identifier and index anchor)
- **RAM:** Random Access Memory (Volatile operational system and execution memory)
- **RBAC:** Role-Based Access Control (Permission and privilege enforcement security model)
- **RDBMS:** Relational Database Management System (Structured relational data store)
- **REST:** Representational State Transfer (Stateless HTTP architectural pattern)
- **RFC:** Request for Comments (IETF standard specification and technical memorandum)
- **SDK:** Software Development Kit (Collection of software development frameworks and libraries)
- **SHA:** Secure Hash Algorithm (Cryptographic one-way hash family, e.g., SHA-256)
- **SMTP:** Simple Mail Transfer Protocol (Electronic mail transmission standard protocol)
- **SQL:** Structured Query Language (Domain-specific relational database query language)

- **SSL:** Secure Sockets Layer (Cryptographic transport layer protocol predecessor to TLS)
- **SVG:** Scalable Vector Graphics (XML-based two-dimensional vector graphic format)
- **TCP:** Transmission Control Protocol (Connection-oriented, reliable transport layer protocol)
- **TLS:** Transport Layer Security (Modern cryptographic web communication protocol)
- **UI:** User Interface (Visual presentation and interaction layer for human operators)
- **UML:** Unified Modeling Language (Standardized visual software modeling language)
- **URI:** Uniform Resource Identifier (Compact character sequence identifying abstract resources)
- **URL:** Uniform Resource Locator (Web network address identifier)
- **UTC:** Coordinated Universal Time (Primary global atomic time standard)
- **UX:** User Experience (Overall human interaction, accessibility, and satisfaction paradigm)
- **VM:** Virtual Machine (Virtualization and emulation of computer systems)
- **VPS:** Virtual Private Server (Dedicated virtualized cloud hosting node)
- **W3C:** World Wide Web Consortium (International web standards organization)
- **WGS-84:** World Geodetic System 1984 (Standard global coordinate reference frame)
- **WSGI:** Web Server Gateway Interface (Python synchronous web application server standard)
- **XSS:** Cross-Site Scripting (Client-side malicious code and script injection vulnerability)

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- **Figure 5.1:** Synthesized Canine Health Passport (ReportLab PDF Artifact)
- **Figure 5.2:** Synthesized Legal Canine Breeding Contract (ReportLab PDF Artifact)

Experimental Benchmarks and Performance Charts (Chapter 6)

- **Figure 6.1:** Spatial Proximity Query Latency vs. Canine Listing Volume
- **Figure 6.2:** Geographic Coordinate Resolution Latency Comparison
- **Figure 6.3:** Legal Contract & Passport PDF Compilation Throughput